LIQPLAT Statistical Analysis Plan

2026-01-09

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1 Administrative information

1.1 Revision history

Version	Date	Who	Comments
0.1	2024-09-20	Johannes Schwenke	First draft
0.9	2025-10-27	Johannes Schwenke	Consolidated draft based on results from simulations studies

1.2 Roles and responsibilities

Name	Affiliation	Role
Johannes Schwenke	University of Basel and University Hospital of Basel	Trial Statistician
Giusi Moffa	Independent	Advising Statistician
Benjamin Kasenda	University Hospital of Basel	Principal investigator and Sponsor-Investigator
Matthias Briel	University of Basel and University Hospital of Basel	Methodologist

1.3 Abbreviations

Abbreviation	Full Term
ATE	Average Treatment Effect
CDWH	Clinical Data Warehouse
CrI	Credible Interval
ctDNA	Circulating Tumor DNA
DAG	Directed Acyclic Graph
ECOG	Eastern Cooperative Oncology Group
EORTC	European Organisation for Research and Treatment of Cancer
GRC	General Research Consent
HR	Hazard Ratio
ITT	Intention-to-Treat
KM	Kaplan-Meier
MAR	Missing At Random
MCAR	Missing Completely at Random
MCMC	Markov Chain Monte Carlo
MICE	Multiple Imputation by Chained Equations
OR	Odds Ratio
OS	Overall Survival
PFS	Progression-Free Survival

Abbreviation	Full Term
PMM	Predictive Mean Matching
PROMs	Patient Reported Outcome Measures
psATE	Principal Stratum Average Treatment Effect
QoL	Quality of Life
RMST	Restricted Mean Survival Time
SAP	Statistical Analysis Plan
SAT	Single-Arm Trial
SOP	State Occupancy Probability
TAOOH	Time Alive and Out of Hospital

1.4 Signatures

This Statistical Analysis Plan has been reviewed and approved by the undersigned:

Principal Investigator

Name: Benjamin Kasenda

Signature: ____________________________________

Date: __________________________________

Trial Statistician

Name: Johannes Schwenke

Signature: ____________________________________

Date: __________________________________

2 Introduction

The purpose of this Statistical Analysis Plan (SAP) is to provide a detailed description of the intended statistical analyses for the LIQPLAT trial. The SAP is based on the protocol version 1.1.

We developed this SAP using data sets based on routinely collected data of patients who sought care prior to LIQPLAT, and who would have fulfilled the eligibility criteria. These data are complemented with simulations where needed.

3 Trial Overview

3.1 Background and rationale

Liquid biopsies, particularly circulating tumor DNA (ctDNA) analysis, offer potential solutions to challenges in treating patients with solid cancers. ctDNA may provide a more comprehensive view of tumor heterogeneity than traditional biopsies, aiding in identifying targetable alterations, quantifying disease burden, detecting early treatment resistance, and predicting outcomes. While promising, more robust trial data on the implementation of ctDNA in routine care is needed. We therefore conduct a single-arm trial (SAT) to investigate ctDNA implementation in routine care for advanced cancer patients, using random invitation to ensure representative sampling. The random invitation to participation allows for a randomized comparison with eligible patients not invited to the trial.

3.2 Objectives

As per protocol version 1.1 LIQPLAT has two primary objectives:

1. To assess the implementation and feasibility of ctDNA measurements from peripheral blood during routine clinical care of cancer patients in the University Hospital Basel.
2. To compare clinical and patient-reported outcomes of patients from this trial with patients who did not receive systematic ctDNA measurement.

This SAP focuses primarily on the second objective, detailing the analyses for the three co-primary outcomes of the randomized comparison: Overall Survival (OS), Quality of Life (QoL), and Time Out of Hospital (TAOOH).

4 Study methods

4.1 Trial design

LIQPLAT is a single-arm, single-center trial at the University Hospital Basel. Patients are randomly invited from an ongoing research registry for participation in the trial (Figure 1). The ongoing research registry is made up of all patients who have signed the general research consent (GRC) of the University Hospital Basel.

Based on past-data of the Division of Medical Oncology, around 240-300 patients would be eligible for the trial over a period of 18 months. However, for logistic reasons and capacity of the Department of Pathology, we limit the sample size to 150 participants, with a maximum of three patients selected for invitation to the trial per week.

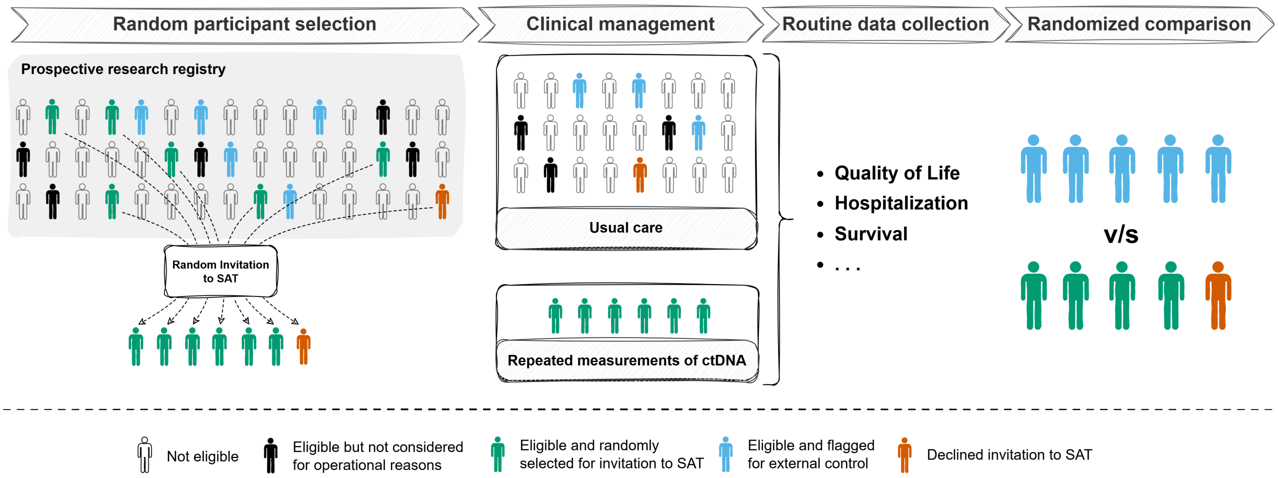


Figure 1: Patients are randomly selected for invitation to the single-arm trial from a prospective research registry. Some eligible participants might not be considered for selection for operational reasons, e.g., inpatients who require emergency treatment before trial staff was made aware of their existence. All outcome data are obtained from routinely collected data. The design enables an unconfounded (randomized) comparison of patients who were selected for invitation to the single-arm trial with patients who were eligible but not selected for invitation.

4.1.1 Trial population

4.1.1.1 Inclusion criteria

1. All adult patients who are part of the University Hospital Basel prospective research registry, i.e., have signed the GRC, with
2. A proven advanced solid malignant disease, and
3. An indication for medical anti-cancer treatment (including combined chemo-radiotherapy or immuno-radiotherapy).

4.1.1.2 Exclusion criteria

1. Patients with primary brain tumors,
2. Patients with primary resectable disease, and
3. Patients with prior treatment for advanced or metastatic disease.

4.1.2 Random Invitation to the Single-Arm Trial

4.1.2.1 Screening list

We build a weekly screening list containing all patients who have an appointment for a first consultation for advanced cancer at the Division of Medical Oncology in the following week in REDCap (v. 14.3.14), using routinely collected data from the electronic patient records. The list is randomly ordered by sorting a hashed version (SHA-256) of the sequentially generated patient hospital ID. The trial team (at least one board-certified oncologist) checks whether patients fulfill the eligibility criteria every Friday afternoon.

4.1.2.2 Invitation to the Single-Arm Trial

Descending the screening list row by row, we select up to three patients per week who will be invited to the trial. The decision to invite is made randomly, using a random sampling table uploaded to REDCap. The random sampling table was generated by a trialist not associated with the trial using R version 4.3.1 (2023-06-16) R Core Team (2025). The table was stratified by presence or absence of a primary diagnosis of a lung cancer, with a block size of 9, and has odds of 2:1 for an invitation to the trial. All investigators are blinded to the random sampling table.

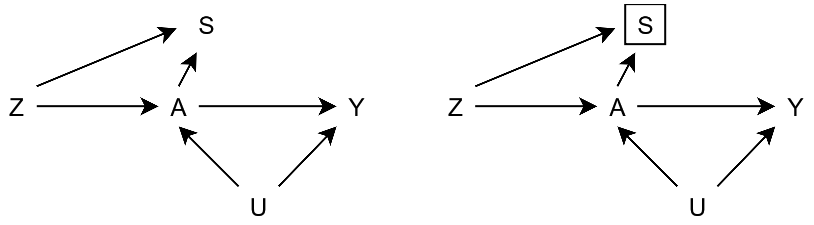
Once three patients have been randomly selected for an invitation to the trial, the list is closed for the current week and a new one is started the next week. Inpatients can also be invited when fulfilling eligibility criteria and slots are available for the given week.

4.1.3 Randomized comparison with external comparator

Randomly inviting patients from a prospective research registry to the SAT enables a randomized comparison with patients who were not invited to the trial, as the data on outcomes for the latter can be obtained from routine data. In Figure 2 Z is the indicator for selection for invitation to the SAT, A indicates whether a patient was invited to the study *and* accepted, S indicates the selection into the population who accepted the invitation, Y denotes the outcome of a patient, and U denotes a set of unmeasured causes A and Y. Selection for nvitation to the trial (Z) and the outcomes of interest (Y) have no common causes, because the decision to select a patient for invitation to the SAT was made at random.

As all patients who are considered for LIQPLAT are part of the prospective research registry and have therefore agreed for their data to be used for research purposes, we can compare the outcomes of patients who were invited to the trial to those who were not invited. However, a patient's decision to accept the invitation to the SAT (A) is not random and likely to be influenced by factors (U) which also cause the outcome. If we restricted the analysis to patients who have accepted the invitation (as represented by a box around S), we would be conditioning on the child of a mediator, leading to collider bias. In other words, patients who were selected for invitation to the SAT and patients who could have been selected but were not are exchangeable regarding their potential outcomes (Rubin 2005), but participants of the SAT are not exchangeable with non-participants. This means that for the randomized comparison, the random selection for invitation to the SAT is considered as the treatment.

Even though LIQPLAT is not a randomized controlled trial, when considering selection for invitation to the trial as treatment, it can be analyzed analogously to a randomized controlled trial. We will therefore specify comparative analyses in line with the estimand framework (Kahan et al. 2024), where we consider invitation to the SAT as the treatment. An analysis of the effect of actual participation in the SAT, would require additional assumptions, which are outlined in Section 13.1, but are not part of this SAP.



(a) Dag 1 (b) Dag 2

Figure 2: Simplified causal diagrams for the LIQPLAT trial. Z represents the indicator for selection for invitation to the SAT, A indicates whether a patient was invited and accepted, and Y denotes the outcome. U represents a set of unmeasured common causes of A and Y. S indicates selection into the population who accepted the invitation. While Z and Y have no common causes, A is influenced by factors (U) that also cause the outcome. Restricting the analysis to patients who accepted the invitation —represented by the box around S in (b)— conditions on the child of a mediator, leading to collider bias.

4.1.4 Outcome data

All participants of LIQPLAT are part of the University Hospital's prospective research registry and have thus consented to further use of their routinely collected data for research purposes. We will obtain outcome data almost exclusively from routinely collected data from electronic health records, which are mirrored in the clinical data warehouse (CDWH), and REDCap databases of the Division of Medical Oncology.

Trial specific data, such as dates of random selection, are stored within a REDCap database specific to LIQPLAT.

4.2 Primary Endpoints for Randomized Comparison

This SAP focuses on the three primary endpoints for the randomized comparison between the group selected for invitation to the SAT and the external comparator (not selected for invitation).

1. **Overall Survival (OS):** Time from date of random selection until death due to any cause within 6 months after random selection.
2. **Quality of Life (QoL):** Patient-reported global quality of life measured longitudinally using EORTC questionnaires during routine care up to 6 months after random selection.
3. **Time Alive and Out of Hospital (TAOOH):** Time spent alive and out of hospital up to 6 months after random selection.

Secondary endpoints as described in the protocol (e.g., Progression-Free Survival [PFS], number of imaging exams) will be analyzed but are considered exploratory and are thus no pre-specified in more detail.

4.3 Analysis Populations

4.3.1 Intention-to-Treat (ITT) Population

The ITT population includes all patients from the prospective research registry who were eligible for LIQPLAT according to the criteria in Section 4.1.1 and were considered for random invitation. It comprises two groups defined by the random selection process:

1. **SAT Group (Selected for Invitation):** All eligible patients randomly selected for invitation to participate in the LIQPLAT SAT.
2. **External Comparator (Not Selected for Invitation):** All eligible patients who were considered for invitation but not randomly selected during the same recruitment period.

This population forms the basis for the primary analyses of overall survival and time alive and out of hospital.

4.3.2 Principal Stratum Population

The primary analysis for the QoL endpoint will be conducted on the Principal Stratum Population. This population is defined as the subset of ITT patients who would have survived the entire 26-week (6-month) follow-up period, regardless of their assignment to be invited to the SAT or not. In practice, this population will be approximated by including all patients from the ITT population who were observed to be alive at the end of the 26-week follow-up. The rationale for restricting the QoL analysis to this Principal Stratum is detailed in Section 7.2.3.3.

5 General Statistical Considerations

5.1 Framework

All primary analyses will be conducted within a Bayesian framework. The inferential goal is to estimate the full posterior distribution for the primary estimand for each outcome. This allows for direct probabilistic statements about the plausible magnitude and direction of the treatment effects. We will report key summaries of the posterior distributions (e.g., median, 95% credible interval [CrI]) and relevant probabilities (e.g., probability of benefit, $P(\tau > 0)$ or $P(\tau < 0)$ depending on endpoint direction). We will not pre-specify thresholds for dichotomous claims of success or failure based on these probabilities.

5.2 Missing Data Handling

Missing data are expected for baseline covariates, longitudinal QoL states, and exceptionally for TAOOH states. Death is unlikely to be missing, as the data will be obtained from a very complete registry. We will use Multiple Imputation by Chained Equations (MICE), assuming data are Missing At Random (MAR). Based on

Mathur, Zhang, and Shpitser (2025) we restrict the imputation model to variables included in the substantative analysis or those acting causally on the missing variables or on their missingness indicator, to avoid bias. Imputations will be run using a maximum of 50 iterations for each imputed dataset. We will create $m = 50$ completed datasets.

5.2.1 Longitudinal Outcomes

For QoL (details in Section 13.2) and TAOOH (details in Section 13.3) we will use multilevel imputation, with the patient identifier as the cluster variable, using Predictive Mean Matching (PMM) for all variables.

Level 2 (Patient-Level Imputation)

Baseline variables (e.g., functional status at baseline [Eastern Cooperative Oncology Group {ECOG}], baseline laboratory values), will be imputed using `2lonly.pmm`. This method aggregates longitudinal (Level 1) information, such as daily hospitalization status or QoL by taking the cluster (patient) mean. A linear regression model is fitted to the collapsed data.

$$Y_j = \beta_0 + \mathbf{X}_j\boldsymbol{\beta_1} + \bar{\mathbf{Z}}_j\boldsymbol{\beta_2} + \epsilon_j, \quad \epsilon_j \sim N(0, \sigma_\epsilon^2)$$

Where

- Y_j is the missing baseline variable for patient j,
- $\mathbf{X}_j$ represents other baseline predictors with coefficients $\boldsymbol{\beta_1}$, and
- $\bar{\mathbf{Z}}_j$ represents the aggregated means of longitudinal predictors.

Longitudinal variables at Level 1, such as QoL states or hospitalization states, are imputed using `2l.pmm`, which fits a linear mixed-effects model.

$$Y_{ij} = \beta_0 + \mathbf{X}_{ij}\boldsymbol{\beta_1} + u_{0j} + \epsilon_{ij}, \quad u_{0j} \sim N(0, \sigma_u^2), \quad \epsilon_{ij} \sim N(0, \sigma_\epsilon^2)$$

Where

- Y_{ij} is the missing longitudinal variable for the i-th observation of patient j,
- $\mathbf{X}_{ij}$ represents fixed effects (both time-varying and static) with coefficients $\boldsymbol{\beta_1}$, and
- $u_{0j} \sim N(0, \sigma_u^2)$ represents the patient-specific random intercept.

For both approaches, the final imputed value is selected using Type 1 PMM, as per default in `mice`, with a donor pool of $k = 5$[1].

5.2.2 Imputation for Suvival Analysis

For OS data will be imputed in wide format. For details see Section 13.4.

5.2.3 Diagnostics and inference

Convergence will be assessed via trace plots, the plausibility of the imputations will be assessed using diagnostics graphs comparing the distribution of observed and imputed data.

Following imputation, each analysis model (OS, QoL, TAOOH) will be fitted to each of the $m = 50$ imputed datasets. The posterior draws for all parameters and derived estimands from the 50 analyses will be combined (stacked) to form the final posterior distribution (Zhou and Reiter 2010).

[1]Distance is calculated as $d(i, j) = |\hat{y}_{obs,i} - \dot{y}_{mis,j}|$. Predictions for candidate donors ($\hat{y}_{obs}$) are calculated using the best-fit parameters ($\hat{\beta}$), while predictions for missing cases ($\dot{y}_{mis}$) use stochastic parameters ($\dot{\beta}$) drawn from the posterior distribution (Buuren 2021, chap. 3.4, 7.3, 7.8).

5.3 Software

Analyses will be performed using R (version 4.4.3 or later) (R Core Team 2025). Bayesian models will be fitted using Stan via the `rstanarm` (Goodrich et al. 2024) and `rmsb` (Harrell 2025) packages. Multiple imputation will use the `mice` (van Buuren and Groothuis-Oudshoorn 2011) and `miceadds` (Robitzsch and Grund 2024) packages. For data wrangling and plotting we use the `tidyverse` (Wickham et al. 2019). To assess the convergence of the Markov Chain Monte Carlo (MCMC), we will check $\hat{R}$ for all parameters, with a target value < 1.01, and inspect trace plots to confirm adequate mixing of the chains.

5.4 Model Fitting Strategy

For each analysis model, we will first fit the model without the treatment indicator to assess convergence and evaluate whether model simplification is necessary based on the observed data. Following successful convergence of the reduced model, we will add the treatment indicator and fit the full analysis model.

6 Sample Size

The trial aims to enroll 150 participants who accept the invitation to the single-arm trial (SAT). This target sample size is driven by feasibility, including logistical constraints (such as a maximum of three invitations per week) and the capacity of the Department of Pathology. Based on estimates of patient acceptance, we anticipate needing to invite approximately 180 eligible patients to achieve this target (an acceptance rate of $\approx 83\%$).

The random invitation process, as described in Section 4.1.1, uses a $2:1$ allocation ratio (invitation vs. no invitation). Applying this ratio to the 180 invited patients implies an expected external comparator (not selected for invitation) of approximately 90 patients.

Therefore, the ITT population for the randomized comparison is expected to be $N = 270$. All simulation studies for power and operating characteristics are based on this total sample size of $N = 270$.

This trial is not formally powered for a single primary endpoint using a frequentist framework, as the sample size is fixed by feasibility. Instead, the Bayesian analyses will provide posterior distributions and probabilities of benefit for the three primary endpoints. The operating characteristics presented in the analysis sections provide an indication of the study's ability to detect various plausible effect sizes given this fixed sample size.

The primary estimand for all outcomes is the effect of the treatment policy, which compares the group selected for invitation to the group not selected for invitation. This treatment policy estimand is inherently diluted by non-adherence (i.e., the ≈ 30 patients in the $A = 1$ group who are expected to decline participation).

The simulations (e.g., for TAOOH, Table 8) do not explicitly model this non-uptake mechanism. Instead, the true effect sizes used in the simulations (e.g., a true OR of 0.7) are assumed to represent this already-diluted ITT effect. In other words, a simulated OR of 0.7 reflects the average effect of the invitation strategy across all 180 invited patients (including non-acceptors), compared to the 90 non-invited patients.

7 Co-Primary Outcomes

7.1 Overall Survival

7.1.1 Estimand Definition

The estimand definition for Overall Survival is summarized in Table 1.

Attribute	Definition
Target Population	Adult (≥ 18 years) patients with advanced solid cancer, who have not yet started first line therapy.
Treatment Arms	(A=1): Selection for invitation to the SAT. |
	(A=0): No selection for invitation. |
Variable	Time-to-event: Time (in days) from random selection to death from any cause.
Time Horizon	$t_{max} = 182$ days (6 months).
Summary Measure	Difference in Restricted Mean Survival Time (RMST) at $t_{max} = 182$ days.
Intercurrent Events	See Section 7.1.2.

Table 1: Estimand definition for Overall Survival.

7.1.1.1 Formal Definition

We use the potential outcomes framework (Rubin 2005). Let T_i^a be the potential time-to-death for individual i under treatment assignment $a \in \{0, 1\}$. The survival function under treatment a is $S^a(t) = P(T^a > t)$.

The Restricted Mean Survival Time (RMST) under treatment a up to the time horizon $t_{max} = 182$ days is the area under the survival curve:

$$\text{RMST}^a(t_{max}) = \mathbb{E}[\min\ (T, t_{max})] = \int_0^{t_{max}} S^a(t) dt$$

Our primary estimand for OS is the Average Treatment Effect (ATE) on the RMST at 182 days, denoted τ_{OS}:

$$\tau_{OS} = \text{RMST}^1(t_{max}) - \text{RMST}^0(t_{max})$$

τ_{OS} represents the average difference in days lived within the first 182 days comparing the group selected for invitation to the SAT versus the external control group.

7.1.2 Handling of Intercurrent Events

The handling of intercurrent events for the OS analysis is described in Table 2.

Intercurrent Event	Strategy	Rationale
SAT not offered after selection	Treatment Policy	Analyze as selected (part of A=1). Preserves ITT principle; estimates effect of the *strategy* of offering ctDNA-guided care.
Patient declined participation in SAT	Treatment Policy	Analyze as selected (part of A=1). As above.
Patient changes country / Lost to follow-up	Censoring	Death information is obtained via CDWH / administrative databases, minimizing informative censoring due to death ascertainment. If lost to follow-up, e.g., moved out of country, and known alive at last contact, standard survival censoring applies.
Control patient receives ctDNA test	Treatment Policy	Analyze as not selected (part of A=0). Reflects real-world practice where ctDNA may be used outside the trial context.
Discontinuation of ctDNA monitoring (SAT)	Treatment Policy	Analyze as selected (part of A=1). Adherence does not affect assignment group.

Table 2: Handling of intercurrent events for Overall Survival analysis.

7.1.3 Analysis Method

7.1.3.1 Statistical Model

We use a Bayesian proportional hazards model with an M-spline baseline hazard.

The individual hazard $h_i(t)$ and survival function $S_i(t)$ are given by:

$$S_i(t) = \exp\left[-\int_0^t h_i(u)du\right]$$

$$h_i(t) = h_0(t) \cdot \exp(\eta_i)$$

The baseline hazard $h_0(t)$ is modeled using cubic M-splines with five degrees of freedom[2] on a simplex constraint, to ensure identifiability:

$$h_0(t) = \sum_{k=1}^{5} \gamma_k M_k(t), \quad \text{subject to } \sum_{k=1}^{5} \gamma_k = 1, \gamma_k > 0$$

Given the expected number of events (approximately 50) in the ITT population of $N = 270$ within 6 months, we simplify some predictive covariates. The linear predictor η_i is defined as:

$$\begin{aligned}\eta_i = \beta_0 + \beta_{tx} \cdot \text{tx}_i + \beta_{ecog} \cdot \text{ecog}_i + \beta_{stage} \cdot \text{stage}_i \\ + \sum_{j=1}^{2} \beta_{mGPS,j} \cdot \text{mGPS}_{ij}\end{aligned}$$

where:

[2] In historical data, we compared the following hazard functions: exponential, weibull, B-splines with 5 degrees of freedom (df), M-splines with 5 df, and M-splines with 10 df. The predictive performance of these models according to leave-one-out cross validation did not differ substantially. We therefore chose to model the hazard using M-splines with 5 df, for more flexibility should the trial data be substantially different, but more parsimony than when using 10 df. More details can be found here.

- β_{tx} is the effect of treatment (selection for invitation to the SAT),
- ecog_i is a binary indicator for ECOG performance status (0 if ECOG 0-1, 1 if ECOG ≥ 2),
- stage_i is a binary indicator for disease stage (0 if locally advanced non-operable, 1 if metastatic),
- mGPS_{ij} represents the modified Glasgow Prognostic Score (mGPS) categories, coded as dummy variables for $j \in \{1, 2\}$ (with mGPS=0 as reference).

The modified Glasgow Prognostic Score (mGPS) (Proctor et al. 2011) is a validated prognostic tool based on serum CRP and albumin levels:

- **mGPS = 0:** CRP ≤ 10 mg/L (regardless of albumin)
- **mGPS = 1:** CRP > 10 mg/L and albumin ≥ 35 g/L
- **mGPS = 2:** CRP > 10 mg/L and albumin < 35 g/L

We assign a regularizing prior of $\mathcal{N}(0, 0.55)$ to the treatment effect β_{tx}. This places approximately 80% of the probability mass on hazard ratios between 0.5 and 2.0, reflecting our belief that the intervention is unlikely to more than halve or double the hazard rate. Remaining regression coefficients are assigned default `rstanarm`, weakly informative priors, and the baseline hazard coefficients γ follow a symmetric Dirichlet distribution:

$$
\begin{aligned}
\beta_0 &\sim \mathcal{N}(0, 20) \\
\beta_{tx} &\sim \mathcal{N}(0, 0.55) \\
\beta_{\text{ecog}} &\sim \mathcal{N}(0, 2.5) \\
\beta_{\text{stage}} &\sim \mathcal{N}(0, 2.5) \\
\beta_{\text{mGPS}, j \in 1,2} &\sim \mathcal{N}(0, 2.5) \\
\gamma &\sim \text{Dirichlet }(\mathbf{1}_5)
\end{aligned}
$$

7.1.3.2 Implementation

The model will be fitted using the `stan_surv` function from the `rstanarm` package in R. Convergence will be assessed using $\hat{R}$ and trace plots. Example impleme1000ntation code is provided in Section 13.7.

7.1.3.3 Obtaining the Estimand

Using the fitted model we generate posterior draws of the marginalized survival curves for under each treatment ($a = 0$ and $a = 1$) up to $t_{max} = 182$ days. For each posterior draw, we calculate the RMST under both treatments ($\widehat{\text{RMST}}^0$ and $\widehat{\text{RMST}}^1$) by integrating the respective survival curves from $t = 0$ to $t_{max} = 182$ days. We will approximate the integral using the trapezoidal rule on 200 discrete time points. We then compute the difference $\hat{\tau}_{OS,\text{ draw}} = \widehat{\text{RMST}}^1_{\text{draw}} - \widehat{\text{RMST}}^0_{\text{draw}}$. The collection of $\hat{\tau}_{OS,\text{ draw}}$ values forms the posterior distribution of the primary estimand. Example code for obtaining the RMST difference is provided in Section 13.7.

Besides the RMST, we will also derive risk differences and risk ratios at 6 months from the same model. These will be calculated for each posterior draw by comparing the marginal survival probabilities at $t = 182$.

7.1.3.4 Operating Characteristics

We evaluated the operating characteristics OS endpoint through a simulation study based on historical patient data.

To ensure realistic survival trajectories, we generated data using a second-order Markov ordinal transition model fitted to historical *daily* status data from the Division of Medical Oncology. This model predicts daily transitions between five health states, where State 5 (Death) is an absorbing state. For each simulation iteration, we generated trajectories for $N = 270$ patients, allocated 2:1 to the intervention ($n = 180$) and control ($n = 90$) arms, mirroring the trial's fixed sample size.

We defined a null scenario (H0) and several alternative scenarios (HA) by introducing a treatment effect into the transition model. The treatment effect was implemented as an odds ratio (OR) applied to the daily transitions, reducing the probability of moving to a worse health state (including death).

These transition-based effects were mapped to the primary estimand, the difference in RMST at 6 months:

- **H0 (Null):** True OR = 1.0 (Treatment re-randomized, True RMST difference $\approx$ 0 days).
- **HA (Small):** True OR = 0.9 (True RMST difference $\approx$ 3 days).
- **HA (Medium):** True OR = 0.8 (True RMST difference $\approx$ 6 days).
- **HA (Large):** True OR = 0.7 (True RMST difference $\approx$ 9 days).
- **HA (Very Large):** True OR = 0.6 (True RMST difference $\approx$ 12 days).

We performed 1,000 simulations for each scenario. In each iteration:

1. Daily health states were simulated for all patients up to 182 days.
2. The daily trajectories were reduced to a standard time-to-event format (time to death or censoring at 182 days).
3. The primary analysis model (Bayesian proportional hazards with M-spline baseline hazard) was fitted to the synthetic dataset.
4. Power was calculated as the proportion of simulations where the posterior probability of benefit ($P(\tau_{OS} > 0)$) exceeded 0.95.

True difference in RMST (days)	Assertion Probability (95% CI)
0.0	2.9% (1.4% - 4.3%)
3.1	12.2% (9.3% - 15.2%)
6.5	32.2% (28.1% - 36.4%)
9.2	59.0% (54.6% - 63.3%)
11.6	84.9% (81.7% - 88.1%)

Table 3: Operating characteristics for Overall Survival analysis under various true effect sizes (difference in RMST at 182 days).

7.1.3.5 Visualization of Estimate

We will visualize the survival curves and τ_{OS} as shown in Figure 3 and Figure 4.

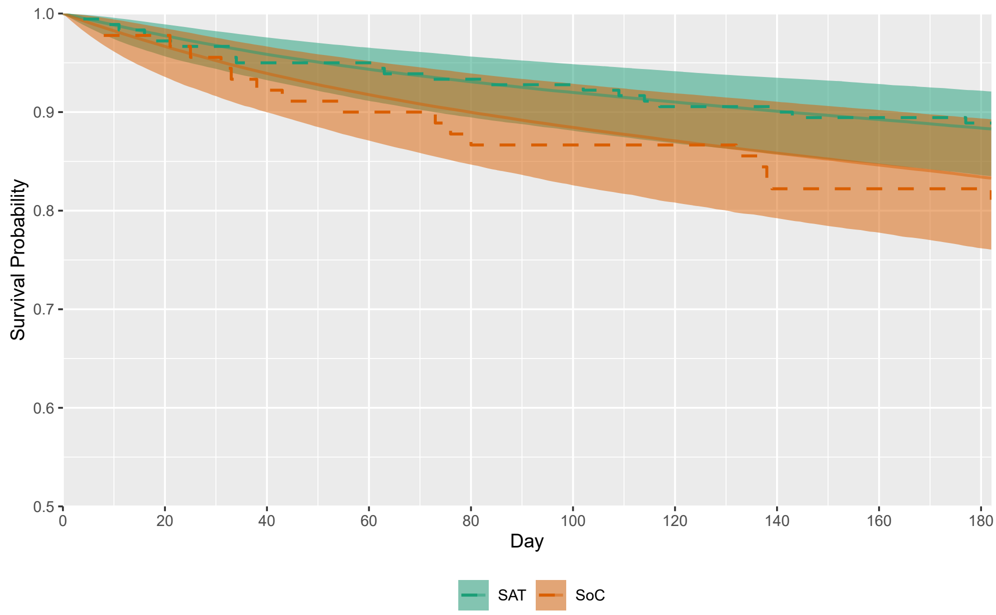


Figure 3: This figure shows the model derived marginal survival curves, with the Kaplan-Meier curves overlaid as dashed lines. The sample ($n = 270$) is from simulated data with a true underlying $OR = 0.9$ for transitioning to a worse state for the group selected for invitation to the single-arm trial ($\tau_{OS} \approx 3$ days). Abbreviations: SAT; single-arm trial, SoC; standard of care.

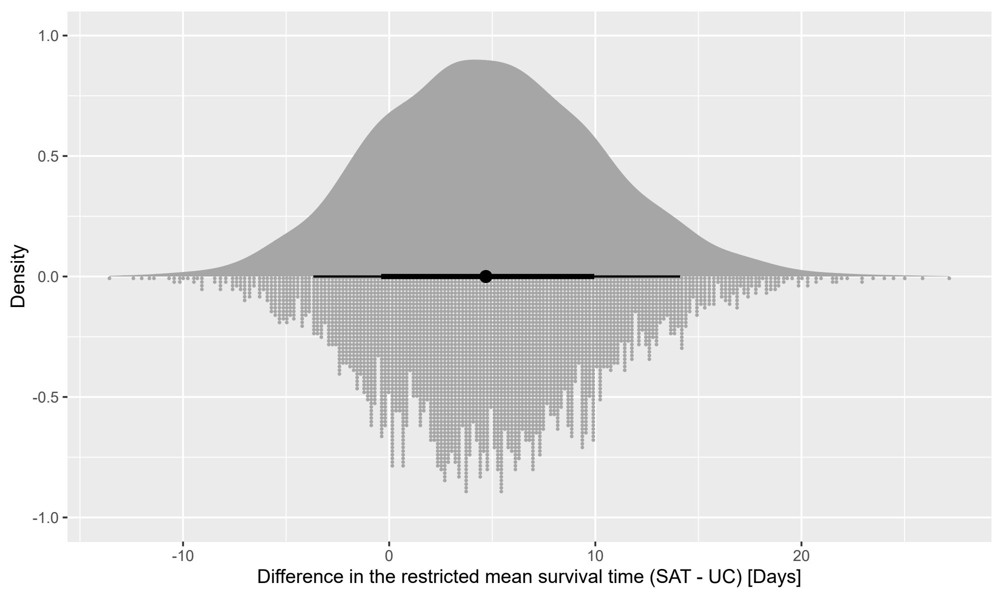


Figure 4: Posterior distribution of the primary estimand, the expected difference the restricted mean survival time ($\hat{\tau}_{OS}$). The sample ($n = 270$) is from simulated data with a true underlying odds ratio of 0.9 for transitioning to a worse state for the group selected for invitation to the single-arm trial ($\tau_{OS} \approx 3$ days). The 66% and 95% credible interval are shown. Abbreviations: SAT; single-arm trial, SoC; standard of care.

7.2 Quality of Life

7.2.1 Data

In LIQPLAT QoL, measured using the European Organisation for Research and Treatment of Cancer (EORTC)

QLQ-C30 and QLQ-C15 questionnaires (Fayers et al. 2002).[3] The QoL scale (question 30) is an ordinal outcome with seven levels, ranging from 1 (very poor) to 7 (excellent), as shown below:

How would you rate your overall quality of life during the past week?

1	2	3	4	5	6	7
Very poor						Excellent

For modeling purposes, we invert the scale. We will also assume that the categories that are not labeled according to EORTC can be labeled as follows:[4]

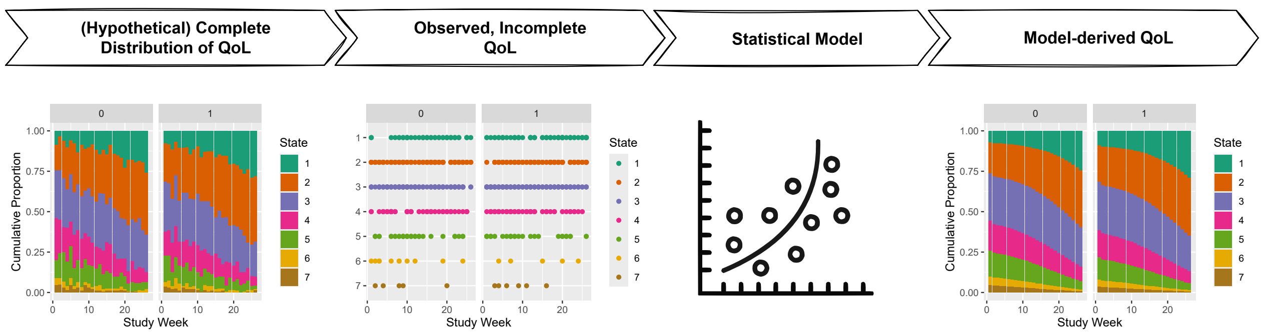
1	2	3	4	5	6	7
Excellent	Very Good	Good	Average	Below Average	Poor	Very poor

In LIQPLAT, QoL questionnaires are completed at clinical appointments, which occur frequently but at irregular intervals for each patient. We anticipate an average interval of about six weeks. This data structure has key implications.

1. **Time-Dependent Correlation:** A patient's QoL at one time point is correlated with their QoL at preceding time points. This within-subject correlation should be accounted for (Harrell 2015). In other words, multiple measurements within a subject are *not* exchangeable, as would be assumed by a simple random intercept and the associated compound symmetric correlation structure (Harrell 2015).
2. **Irregular Gaps:** The correlation between two measurements likely weakens as the time gap between them increases. The analytical model should explicitly incorporate the time gap between consecutive measurements to better approximate the decaying correlation structure. If we assume that the irregularity of follow-up is random conditional on a set of baseline covariates and the previous QoL state, the irregular spacing could be used to reconstruct QoL trajectories over the entire follow-up period.
3. **Competing Risk:** In a population with advanced cancer, death is a frequent outcome that prevents any further QoL measurement. This represents a competing risk. As detailed in Section 7.2.3.3 this competing risk is a major challenge.

7.2.2 General Approach

We propose to make use of the irregular, near-random follow-up of usual care and its QoL measurements to reconstruct the QoL trajectories for the entire duration of follow-up (see Figure 5).



[3]The EORTC's scoring manual advocates for treating the responses as numeric (Fayers et al. 2001), which we will not do. The shortcomings of treating ordinal scores as if they were numeric have explained in detail elsewhere (Liddell and Kruschke 2018).

[4]We acknowledge that not everyone would assign precisely these labels, but believe them to be agreeable to the majority of patients. The labels will serve for easier communication of results.

Figure 5: Conceptual overview of the statistical approach for reconstructing Quality of Life trajectories. The analysis bridges the gap between the hypothetical, complete QoL data (far left), from which empirical state occupancy probabilities could be directly calculated, and the actual observed data, which is sparse and irregularly spaced (second from left). A statistical model (third from left) is fitted to these sparse observations to generate complete, model-derived SOPs for the entire follow-up period (far right), reconstructing the full QoL trajectory for the population and providing associated uncertainty estimates.

7.2.3 Estimand Definition

The estimand definition for Quality of Life is summarized in Table 4.

Attribute	Definition
Target Population	The Principal Stratum of adult (≥ 18 years) patients with advanced solid cancer, who have not yet started first-line therapy, defined as patients who would survive at least 26 weeks after their first appointment.
Treatment Arms	(A=1): Selection for invitation to the SAT. |
	(A=0): No selection for invitation to the SAT. |
Variable	Ordinal QoL state (7 levels, derived from EORTC QLQ-C30 Q30) measured longitudinally at irregular intervals. We use an inverted scale where 1='Excellent', ..., 7='Very Poor'.
Time Horizon	$t_{max} = 26$ weeks (6 months).
Summary Measure	Difference in mean number of weeks spent with "Good" QoL (defined as states 1, 2, or 3) during the 26-week period, within the Principal Stratum.
Intercurrent Events	See Section 7.2.4.

Table 4: Estimand definition for Quality of Life.

7.2.3.1 Formal Definition

Let S_i^a be an indicator for survival of patient i at 26 weeks under treatment assignment $a \in \{0, 1\}$. $S_i^a = 1$ if patient i survives, $S_i^a = 0$ otherwise. The Principal Stratum of "always-survivors" is the sub-population of patients for whom $S_i^1 = S_i^0 = 1$.

Let Y_{it}^a be the potential ordinal QoL state (1-7, 1=best) for patient i at week t under treatment a. Let W_i^a be the potential total number of weeks patient i spends in a 'good' or better QoL state ($j \in \{1, 2, 3\}$) under treatment a over 26 weeks:

$$W_i^a = \sum_{t=1}^{26} \sum_{j=1}^{3} \mathbb{I}(Y_{it}^a = j)$$

The primary estimand for QoL, the Principal Stratum Average Treatment Effect (psATE) on weeks in a "Good" state, denoted $\tau_{QoL}^{\text{psATE}}$, is:

$$\tau_{QoL}^{\text{psATE}} = \mathbb{E}[W_i^1 \mid S_i^1 = 1, S_i^0 = 1] - \mathbb{E}[W_i^0 \mid S_i^1 = 1, S_i^0 = 1]$$

7.2.3.2 Identification Assumption

Identification of $\tau_{QoL}^{\text{psATE}}$ by analyzing only those patients *observed* to survive 26 weeks relies on the key assumption that selection for invitation to the SAT has no effect on 26-week survival status: For all patients i, $S_i^1 = S_i^0$. Under this assumption, the group observed to survive is equivalent to the "always-survivor" Principal Stratum.

7.2.3.3 Rationale for Principal Stratum

Modeling the QoL trajectory while incorporating the absorbing state of death proved problematic with sparse intermediate QoL measurements common in routine care. Our simulations indicated that interval-censoring approaches for missing QoL states led to biased estimates, particularly overestimating the probability of death (Section 13.5). Assuming no treatment effect on 6-month survival allows us to focus the QoL analysis on the stratum of patients who survive this period regardless of treatment, providing a potentially less biased estimate of the QoL effect *for this subgroup.*

We will formally assess the assumption of no survival difference using the results from the Overall Survival analysis. If the 66% CrI for the difference in Restricted Mean Survival Time (RMST) at 6 months, τ_{OS}, is contained within a Region of Practical Equivalence (ROPE) of ± 7 days (i.e., $[-7, 7]$ days), we will consider the assumption of no survival difference to be practically met and proceed with the Principal Stratum estimand as the primary analysis.

If the 66% CrI for τ_{OS} is not contained within the ROPE we will switch the primary analysis to the ordinal QoL state at 6 months, adjusted for death. In this case, death will be assigned the worst possible QoL score (state 8), and the outcome will be analyzed using a Bayesian proportional odds model (see Section 8). Additionally, we will report the results of the longitudinal ordinal transition model that includes death as an absorbing state (see Section 8) to describe the treatment effect on the combined outcome of survival and quality of life over time.

7.2.4 Handling of Intercurrent Events

The handling of intercurrent events for the QoL analysis is described in Table 5.

Intercurrent Event	Strategy	Rationale
Death	*Principal Stratum*	Death before 26 weeks defines exclusion from the primary analysis population. This relies on the assumption in Section 7.2.3.2.
Discontinuation of ctDNA monitoring (SAT)	*Treatment Policy*	Analyze as selected (part of A=1). Preserves ITT principle; estimates the effect of the *strategy* of offering ctDNA-guided care, irrespective of adherence.
Switch to best supportive care	*Treatment Policy*	Patient remains in their assigned group (A=1 or A=0). Patients generally remain under the care of the Division of Medical Oncology, although visit frequency may decrease. The longitudinal transition model handles irregular intervals and projects QoL states for unobserved weeks based on the last observed state and model parameters (MAR).
Treatment at another hospital	*Treatment Policy*	Patient remains in their assigned group (A=1 or A=0). We project the QoL trajectory for the unobserved period using the transition model parameters estimated from the available data (MAR).
SAT not offered after selection	*Treatment Policy*	Analyze as selected (part of A=1).
Patient declined participation in SAT	*Treatment Policy*	Analyze as selected (part of A=1).

Intercurrent Event	Strategy	Rationale
Control patient receives ctDNA test	*Treatment Policy*	Analyze as not selected (part of A=0).

Table 5: Handling of intercurrent events for Quality of Life analysis.

⚠ Warning

Informative Missingness due to Routine Care Data Collection: A design limitation is that QoL is measured during routine appointments. If the ctDNA strategy influences clinical decisions (e.g., earlier switch to best supportive care , different follow-up intensity), it affects *both* the patient's true QoL *and* the probability of observing QoL. This creates a potential for informative missingness. To assess the potential for differential informative missingness, we will compare the cumulative incidence of switching to best supportive care between the two trial arms. If the rates and timing of switching to best supportive care are similar between groups, it reduces the concern that missingness driven by palliative care transitions biases the treatment comparison.

7.2.5 Analysis Method

To target the estimand we use a first-order Markov longitudinal ordinal transition model (Rohde et al. 2024). This model analyzes the probability of transitioning between QoL states from one week to the next, conditional on the previous state and the time gap since the last measurement. This approach accounts for the ordinal nature, longitudinal correlation, irregular measurement times, and provides a framework to derive interpretable summary measures like time spent in specific states. A higher-order Markov model would not be feasible, as quality of life states prior to baseline are not available.

7.2.5.1 Statistical Model

Let y_{it} be the ordinal QoL state (1-7, 1=best) for patient i at week t. Let $y_{it'}$ be the last observed state at a prior week t', and let the time gap be $\Delta t = t - t'$. The model is a cumulative logit model for transition probabilities:

$$\operatorname{logit}\left(P(y_{it} \geq j \mid y_{it'}, t, \Delta t, \mathbf{X}_i, \mathrm{tx}_i)\right) = \alpha_j + \eta_{it} + \gamma_{it,j}, \quad \text{for} \quad j = 2, \ldots, 7$$

The category-specific intercepts α_j define the baseline cumulative probabilities of being in state j or worse. The linear predictor η_{it} captures the effects that are assumed to satisfy the proportional odds assumption:

$$\begin{aligned}\eta_{it} = \beta_{\mathrm{tx}} \cdot \mathrm{tx}_i + f(t) + \sum_{k=2}^{7} \beta_{\mathrm{yprev}\ =k} \cdot \mathbb{I}(y_{it'} = k) \\ + \beta_{\mathrm{gap}} \cdot \Delta t + \sum_{k=2}^{7} \beta_{\mathrm{yprev}\ =k \times\ \mathrm{gap}} \cdot \mathbb{I}(y_{it'} = k) \cdot \Delta t \\ + \sum_{j=2}^{3} \beta_{ecog,j} \cdot \mathrm{ecog}_{ij} + \sum_{k=2}^{5} \beta_{diag,k} \cdot \mathrm{diag}_{ik}\end{aligned}$$

where β_{tx} is the main effect of treatment selection (A=1 vs A=0),[5] and categorical predictors (ecog, diag) are

[5] A treatment effect, if present, is unlikely to be constant over time. CtDNA analysis can take weeks and is less likely to change first-line than second-line therapy of participants. An effect in the first weeks after baseline is thus very unlikely. However, given our sample size, we are likely not able to identify time-varying treatment effects reliably and choose a constant treatment effect for slightly smaller variance in our estimator.

represented by sums over their dummy variables. We model time flexibly[6] using a restricted cubic spline with 4 knots,[7] which generates 3 basis functions:

$$f(t) = \sum_{p=1}^{3} \beta_{t,p} \cdot B_p(t)$$

The effect of the previous QoL state ($\beta_{\text{yprev }=k}$, with state 1 as reference)[8] and its interaction with the time gap ($\beta_{\text{yprev }=k\times \text{ gap}}$) capture how the current state depends on the previous measurement and the time elapsed since then.[9] The baseline covariates $\mathbf{X}_i\boldsymbol{\beta}_x$ include ECOG performance status and diagnosis category.[10]

The term $\gamma_{it,j}$ models deviation from proportional odds for the effect of time, constrained to be linear in the outcome category j:

$$\gamma_{it,j} = (\tau \cdot t) \cdot j$$

where τ is the non-proportional odds parameter for time.

7.2.5.2 Prior Specifications

We assign a regularizing prior of $\mathcal{N}(0, 0.5)$ to the treatment effect β_{tx}. We used prior predictive simulation to select this prior, which implies a clinically plausible distribution for our estimand $\tau_{QoL}^{\text{psATE}}$. Our simulation, based on historical data, suggests that this prior on the log odds scale[11] produces a prior distribution for $\tau_{QoL}^{\text{psATE}}$ with a median at 0 weeks (no difference), with approximately 80% of its mass on an increase in weeks with "good" or better QoL by 4 weeks (benefit) to a decrease by 3 weeks (harm).

The intercepts α_j have priors induced by a Dirichlet distribution with a concentration parameter of 0.308[12] on the baseline cell probabilities, which ensures proper ordering ($\alpha_2 < \ldots < \alpha_7$).[13] All other regression coefficients—including the time spline coefficients, previous state effects, gap effects, their interactions, and

[6]The analysis discretizes time into weekly intervals rather than daily intervals for several pragmatic and methodological reasons. First, in the EORTC questionnaire patients are specifically asked about their quality of life in the last week. Second, given the high proportion of missing data inherent to routine care follow-up, daily estimates could imply a level of precision that is not supported by the data. Third, in LIQPLAT, it is highly unlikely that patients complete more than one questionnaire within a single week, making the week a natural and minimal unit of observation. Finally, this choice significantly reduces the computational burden of the simulation study and the primary analysis (e.g., 26 transitions vs. 182 transitions per patient), enhancing the feasibility of the project without a substantial loss of relevant information. The impact of this aggregation on the final estimand is expected to be minimal.

[7]This model uses the `rmsb` package, which implements restricted cubic splines using a truncated power basis. This basis is parameterized for convenient interpretation: for $K = 4$ knots, it generates $K - 1 = 3$ basis functions ($B_p(t)$). The first basis function ($B_1(t)$) is the linear term t itself. The subsequent functions ($B_2(t)$,$B_3(t)$) are the nonlinear components derived from truncated cubic functions. Therefore, the coefficients $\beta_{t,1}$, $\beta_{t,2}$, $\beta_{t,3}$ are the weights for the linear component and the two nonlinear components, respectively.

[8]It is unlikely that the effect of the previous state is constant over time. Very poor QoL at the beginning of the study might predict regression to the mean, whereas a very low-quality of life during later stages of the disease might not. However, estimating an interaction of the six-level categorical variable `yprev` with flexibly modeled time would not be feasible given our sample size.

[9]The effect of the previous state is unlikely to decay linearly over time. However, most questionnaires will be repeated after at least some weeks. Even if the decay shortly after a previous answer is non-linear, we anticipate the decay in the expected range of gaps to be approximately linear.

[10]Both the functional status (ECOG) at baseline and the cancer diagnosis category are likely to be predictive of quality of life based on our clinical experience and prior research, and are thus included to explain more variation in the outcome. If we observe fewer than five patients with a functional status (ECOG) of 4, we will merge categories 3 and 4 into '3plus', to avoid divergent transitions of the estimate of the effect of ecog 4 because of too few observations. The diagnosis categories will be refactored into a five level categorical variable, with the four most common diagnoses as distinct categories and the rest of diagnoses lumped together as 'other'.

[11]In other words, we place 80% of our prior probability mass for the odds ratio of the treatment effect between 0.53 and 1.90.

[12]The concentration parameter as calculated by default by the `rmsb` package is $1/(0.8 + 0.35 \cdot \max\ (k, 3))$, which evaluates to 0.308 for a 7-level outcome.

[13]Unlike the TAOOH model, we do not include a patient-level random intercept. With an anticipated average of only ~4 observations per patient for this outcome, estimation of a random intercept variance component is expected to be unstable.

baseline covariate effects—are assigned weakly informative $\mathcal{N}(0, 100)$ priors (the default in `rmsb`). The non-proportional odds parameter τ also receives a $\mathcal{N}(0, 100)$ prior.

$$
\begin{aligned}
\pi_1, ..., \pi_K &\sim \text{Dirichlet }(0.308)\text{ , which induces priors on } \alpha_j \\
\beta_{tx} &\sim \mathcal{N}(0, 0.5) \\
\beta_{t,p\in 1,2,3} &\sim \mathcal{N}(0, 100) \\
\beta_{gap} &\sim \mathcal{N}(0, 100) \\
\beta_{\text{yprev }=k\in 2..7} &\sim \mathcal{N}(0, 100) \\
\beta_{\text{yprev }=k\times\text{ gap }\in 2..7} &\sim \mathcal{N}(0, 100) \\
\beta_{ecog,j\in 2,3} &\sim \mathcal{N}(0, 100) \\
\beta_{diag,k\in 2..5} &\sim \mathcal{N}(0, 100) \\
\tau &\sim \mathcal{N}(0, 100)
\end{aligned}
$$

7.2.5.3 Implementation

The model will be fitted using the `blrm` function from the `rmsb` package in R, restricted to the Principal Stratum population (survivors at 26 weeks). MCMC settings as per OS analysis. Example implementation code is provided in Section 13.8.

7.2.5.4 Obtaining the Estimand

Using the fitted model, we generate posterior draws of the marginalized SOPs under each treatment ($a = 0$ and $a = 1$) over the 26-week follow-up period. For each posterior draw, we calculate $\hat{P}(Y_t^a = j)$ for each week $t = 1, ..., 26$, for each state $j = 1, ..., 7$, and for each treatment. This involves marginalizing over the distribution of baseline covariates, baseline QoL states, and the Markov process transitions over time.

For each posterior draw, we then calculate the expected number of weeks in a "Good" state (1, 2, or 3) for each treatment arm by summing the relevant SOPs over the 26 weeks:

$$
\hat{\mathbb{E}}[W^a]_{\text{draw}} = \sum_{t=1}^{26}\sum_{j=1}^{3} \widehat{\text{SOP}}(Y_t^a = j)_{\text{draw}}
$$

We then compute the difference $\hat{\tau}_{W,\text{ draw}}^{\text{psATE}} = \hat{\mathbb{E}}[W^1]_{\text{draw}} - \hat{\mathbb{E}}[W^0]_{\text{draw}}$. The collection of $\hat{\tau}_{W,\text{ draw}}^{\text{psATE}}$ values forms the posterior distribution of the primary estimand for QoL. Example code for obtaining the QoL difference is provided in Section 13.8.

While this omission risks Type I error inflation if significant unmodelled correlation exists, it is considered a necessary trade-off given the sparse data. The Type I error inflation is mitigated by the skeptical prior.

7.2.5.5 Assumptions

! Important

Key assumptions

1. QoL responses can be accurately modeled by using a first-order Markov process. (+)
2. If the effect of time is non-proportional, this can be well approximated using a linear constraint. (+)
3. A quality of life response from a single day is valid for an entire week. (+/-)
4. The treatment effect is proportional, i.e., the odds of changing to a higher category are the same for all thresholds. (+/-)
5. Participants who died during the 6 month follow-up would have died irrespective of their treatment assignment. (+/-)
6. How a previous state affects the current state does not change over time since randomization. (-)
7. The treatment effect, if present, does not change over time. (-)
8. Days where no QoL was assessed are missing at random (MAR). (-)

(+) Indicates that we are confident the assumption is valid, (+/-) indicates that we are uncertain regarding its validity, (-) indicates that we believe the assumption to be strenuous.

Given these assumptions, especially assumptions six to eight, we have **low confidence** in the results. Nevertheless, we believe that other analysis options we have explored are less optimal and would lead us to have very low confidence.

7.2.5.6 Visualization of Estimate

We will visualize the SOPs and τ_{QoL} as shown below.

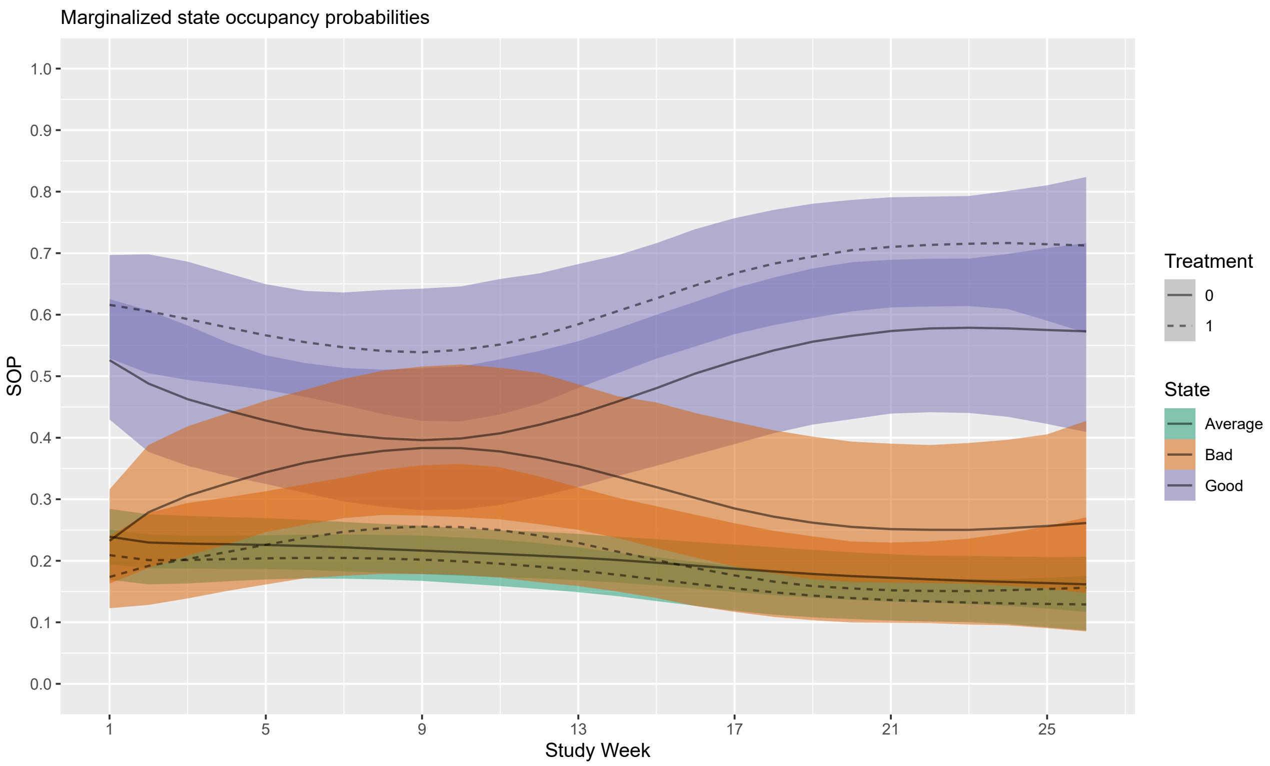


Figure 6: Marginal state occupancy probabilities with 95% credible bands based the model fitted to a sample ($n = 216$) from simulated data with a true difference in good or better Quality of Life of 2 weeks. State

occupancy probabilities have been grouped for illustrative purposes. States 1 to 3 are categorized as 'Good', state 4 as 'Average', and states 5-7 as 'Bad' Quality of Life.

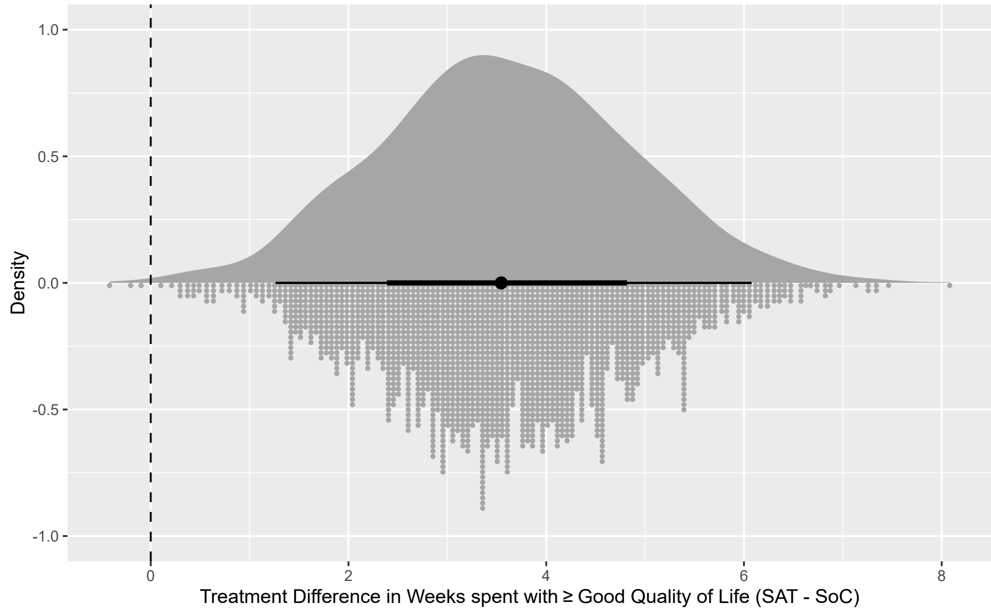


Figure 7: Posterior distribution of the estimated difference in weeks with good or better quality of life ($\hat{\tau}_{QoL}$), defined as state 1, 2, or 3, based on a sample ($n = 216$) of simulated data with a true underlying difference of two weeks. Larger values favour the group selected for invitation to the SAT.

7.2.5.7 Operating Characteristics

Table 6 shows frequentist operating characteristics of the transition model for QoL. The data-generating mechanism assumes a sample size of $N = 216$, as we assume that 20% of initially 270 patients would die before 6 months. 85% of post-baseline observations are missing completely at random (MCAR), which equals about 4 post-baseline measurements per patient. A first order transition model was fit to historical data to obtain plausible QoL trajectories. The simulated data were then generated using a latent Brownian Motion model to approximate the trajectories from the previous step to avoid a perfect match of the data generating model and analysis model. We simulated three scenarios with a difference in good QoL states of 0, 1 week, 2 weeks, and 3 weeks over the 26-week period. For each scenario, 1,000 datasets were simulated and analyzed using the Bayesian first-order Markov ordinal transition model (without covariates). The power is estimated as the proportion of simulations where the posterior probability of any benefit exceeds 0.95. In LIQPLAT we will not use the posterior to make a dichotomous claim of benefit. However, for the purpose of evaluating the operating characteristics of our analysis plan, we assess power and Type I error under the hypothetical decision rule of requiring a posterior probability of benefit greater than 0.95.

τ_{QoL}	Assertion Probability (95% CI)
0.0 weeks	7.3% (5.7% - 8.9%)
1.0 weeks	28.1% (25.3% - 30.8%)
2.0 weeks	60.4% (57.3% - 63.4%)
3.0 weeks	90.5% (88.7% - 92.3%)

Table 6: Probability of assertion of benefit when $P(\text{benefit}) > 0.95$ is chosen as a cut-off.

7.3 Time Out of Hospital

7.3.1 Estimand Definition

The estimand definition for Time Alive and Out of Hospital is summarized in Table 7.

Attribute	Definition
Target Population	Adult (≥ 18 years) patients with advanced solid cancer, who have not yet started first-line anti-cancer therapy.
Treatment Arms	(A=1): Selection for invitation to ctDNA-guided care (SAT).
	(A=0): No selection for invitation (External Comparator). |
Variable	Weekly health state: 1=Alive/Out-of-Hospital, 2=Alive/In-Hospital(Planned), 3=Alive/In-Hospital(ER), 4=Alive/In-Hospital(Unplanned), 5=Dead.
Time Horizon	$t_{max} = 26$ weeks (6 months).
Summary Measure	Difference in mean number of weeks spent Alive and Out of Hospital (State 1) during the 26-week period.
Intercurrent Events	Primarily death, which is modeled as the absorbing state (State 5). See Section 7.1.2 for considerations.

Table 7: Estimand definition for Time Alive and Out of Hospital.

7.3.1.1 Formal Definition

Let Y_{it}^a be the potential health state (1-5) for individual i at week t under treatment assignment $a \in \{0, 1\}$. State 1 represents "Alive and Out of Hospital".

Let W_i^a be the potential total number of weeks individual i spends alive and out of the hospital (State 1) under treatment a over the 26-week period:

$$W_i^a = \sum_{t=1}^{26} \mathbb{I}(Y_{it}^a = 1)$$

Our primary estimand for TAOOH is the Average Treatment Effect (ATE) on this outcome, denoted τ_{TAOOH}:

$$\tau_{TAOOH} = \mathbb{E}[W_i^1] - \mathbb{E}[W_i^0]$$

τ_{TAOOH} represents the population-average difference in weeks spent alive and out of the hospital over 26 weeks.

7.3.2 Handling of Intercurrent Events

Death is part of the outcome definition and explicitly modeled (see Section 7.3.3). Other intercurrent events follow the Treatment Policy approach as outlined for OS (Section 7.1.2).

7.3.3 Analysis Method

7.3.3.1 Longitudinal Ordinal Outcome

We define a patient's state weekly based on their location and vital status: 1=alive and out of hospital, 2=alive and in hospital (planned admission), 3=alive and in hospital (emergency room), 4=alive and in hospital (unplanned admission), 5=dead. This creates a longitudinal ordinal outcome (with state 5 being absorbing).

7.3.3.2 Statistical Model

We use a Bayesian second-order Markov ordinal transition model with patient-level random effects. This model analyzes the probability of transitioning between health states from one week to the next, conditional on the states at the previous two weeks. The second-order specification captures the empirical finding in historic data that both the immediate prior state and the state two weeks earlier contain predictive information about the current health status.[14]

Let y_{it} be the ordinal health state (1-5) for patient i at week t, and let $y_{i,t-1}$ and $y_{i,t-2}$ be the states at the previous two weeks. The model is a cumulative logit model:

$$\text{logit}\ \left(P\left(y_{it} \geq j \mid y_{i,t-1}, y_{i,t-2}, t, \mathbf{X}_i, \text{tx}_i\right)\right) = \alpha_j + \eta_{it} + \gamma_{0i}, \quad \text{for} \quad j = 2, ..., 5$$

The category-specific intercepts α_j define the baseline cumulative probabilities. The linear predictor η_{it} captures the systematic effects:

$$\begin{aligned}
\eta_{it} = {} & \beta_{\text{tx}} \cdot \text{tx}_i + f_0(t) \\
& + \sum_{k=2}^{4} \beta_{\text{yprev1},k} \cdot \mathbb{I}\left(y_{i,t-1} = k\right) + \sum_{l=2}^{4} \beta_{\text{yprev2},l} \cdot \mathbb{I}\left(y_{i,t-2} = l\right) \\
& + \sum_{k=2}^{4} \beta_{t \times yprev1,k} \cdot t \cdot \mathbb{I}\left(y_{i,t-1} = k\right) + \sum_{l=2}^{4} \beta_{t \times yprev2,l} \cdot t \cdot \mathbb{I}\left(y_{i,t-2} = l\right) \\
& + \sum_{j=2}^{3} \beta_{ecog,j} \cdot \text{ecog}_{ij} + \sum_{k=2}^{5} \beta_{diag,k} \cdot \text{diag}_{ik} + f_1(\text{albumin}_i) + f_2(\text{crp}_i)
\end{aligned}$$

where β_{tx} is the main effect of treatment selection (A=1 vs A=0), and categorical predictors (ecog, diag) are represented by sums over their dummy variables. We model time flexibly using a restricted cubic spline with 4 knots, generating 3 basis functions:

$$f_0(t) = \sum_{p=1}^{3} \beta_{time,p} \cdot B_p(t)$$

The effects of the previous two states ($\beta_{\text{yprev1},k}$ and $\beta_{\text{yprev2},l}$, with state 1 as reference) capture the second-order Markov dependency. The summation stops at 4 because state 5 (death) is absorbing. We include interaction terms allowing the effect of each previous state to change linearly over time. Model comparisons on historical weekly state data showed that including these interactions produced large improvements in fit and captures clinically plausible time-varying effects of recent hospital state.

The continuous baseline covariates albumin and CRP are modeled flexibly using restricted cubic splines with 3 knots (2 basis functions each):

$$\begin{aligned}
f_1(\text{albumin}_i) &= \sum_{m=1}^{2} \beta_{alb,m} \cdot B_{m,\ \text{alb}}(\text{albumin}_i) \\
f_2(\text{crp}_i) &= \sum_{n=1}^{2} \beta_{crp,n} \cdot B_{n,\ \text{crp}}(\text{crp}_i)
\end{aligned}$$

The term γ_{0i} is a patient-specific random intercept to account for within-patient correlation beyond the Markov dependency. Longitudinal data such as weekly states are characterized by within-patient correlation. The second-order Markov model accounts for a portion of this correlation by conditioning on the previous two states, but residual correlation may remain due to unobserved patient-level heterogeneity. Our own simulation studies indicate that such unmodelled correlation can inflate the frequentist Type I error rate. The random intercept captures this residual patient-level correlation, aiming to maintain correct Type I error control.

[14]Detailed model fits, AIC comparisons and diagnostic notes can be found here.

7.3.3.3 Prior Specifications

We assign a regularizing prior of $\mathcal{N}(0, 0.541)$ to the treatment effect β_{tx}. We used prior predictive simulation to select this prior, which implies a clinically plausible distribution for our estimand τ_{TAOOH}. Our simulation, based on historical data, suggested that this prior places 80% of its mass on β_{tx} between $\log(0.5) \approx -0.693$ and $\log(2.0) \approx +0.693$, which maps to an asymmetric 80% prior belief interval for τ_{TAOOH} of approximately 6 weeks fewer (harm) to 3 weeks more (benefit) spent alive and out of hospital.

The intercepts α_j have priors induced by a Dirichlet distribution with a concentration parameter of 0.392 on the baseline cell probabilities. All fixed-effect regression coefficients—including the time spline coefficients, previous state effects, their interactions, and baseline covariate effects—are assigned weakly informative $\mathcal{N}(0, 100)$ priors (the default in `rmsb`). The standard deviation of the random intercepts σ_γ is given a Half-t prior with 4 degrees of freedom and scale 1 (the default in `rmsb`).

$$
\begin{aligned}
\pi_1, ..., \pi_K &\sim \text{Dirichlet } (0.392) \text{ , which induces priors on } \alpha_j \\
\beta_{\text{tx}} &\sim \mathcal{N}(0, 0.541) \\
\beta_{time,p\in 1,2,3} &\sim \mathcal{N}(0, 100) \\
\beta_{\text{yprev1},k\in 2..4} &\sim \mathcal{N}(0, 100) \\
\beta_{\text{yprev2},l\in 2..4} &\sim \mathcal{N}(0, 100) \\
\beta_{t\times yprev1,k\in 2..4} &\sim \mathcal{N}(0, 100) \\
\beta_{t\times yprev2,l\in 2..4} &\sim \mathcal{N}(0, 100) \\
\beta_{\text{ecog},j\in 2,3} &\sim \mathcal{N}(0, 100) \\
\beta_{\text{diag},k\in 2..5} &\sim \mathcal{N}(0, 100) \\
\beta_{alb,m\in 1,2} &\sim \mathcal{N}(0, 100) \\
\beta_{crp,n\in 1,2} &\sim \mathcal{N}(0, 100) \\
\gamma_{0i} &\sim \mathcal{N}\left(0, \sigma_\gamma\right) \\
\sigma_\gamma &\sim \text{Half-t } (df = 4, \mu = 0, \sigma = 1)
\end{aligned}
$$

7.3.3.4 Implementation

The model will be fitted using the `blrm` function from the `rmsb` package, including the `cluster(id)` term for random effects. Example implementation code is provided in Section 13.9.

7.3.3.5 Obtaining the Estimand

Using the fitted model, we generate posterior draws of the marginalized State Occupancy Probabilities (SOPs) under each treatment ($a = 0$ and $a = 1$) over the 26-week follow-up period. For each posterior draw, we calculate $\hat{P}(Y_t^a = j)$ for each week $t = 1, ..., 26$, state $j = 1, ..., 5$, and treatment. This requires a recursive calculation accounting for the second-order dependency and marginalizing over baseline covariates, baseline state pairs (y_{-1}, y_0), and random effects. Details can be found in Section 13.6.

For each posterior draw, we then calculate the expected number of weeks in State 1 (Alive and Out of Hospital) for each treatment arm by summing the SOPs for State 1 over the 26 weeks:

$$
\hat{\mathbb{E}}[W^a]_{\text{draw}} = \sum_{t=1}^{26} \widehat{\text{SOP}}(Y_t^a = 1)_{\text{draw}}
$$

We then compute the difference $\hat{\tau}_{TAOOH,\text{ draw}} = \hat{\mathbb{E}}[W^1]_{\text{draw}} - \hat{\mathbb{E}}[W^0]_{\text{draw}}$. The collection of $\hat{\tau}_{TAOOH,\text{ draw}}$ values forms the posterior distribution of the primary estimand for TAOOH. Example code for obtaining the TAOOH difference is provided in Section 13.9.

7.3.3.6 Model Diagnostics: Assessment of Correlation Structure

To validate that our model adequately captures the complex longitudinal dependency in the TAOOH data, we will perform graphical checks. We will compare the empirical correlation structure of the observed weekly states against the correlation structure of data simulated from the fitted model. This comparison will be visualized using Spearman correlation heatmaps (Figure 8) and variograms (Figure 9). This diagnostic check is possible for the TAOOH outcome because it is measured as a complete weekly series; a similar check is not feasible for the QoL endpoint due to its sparse and irregular data collection.

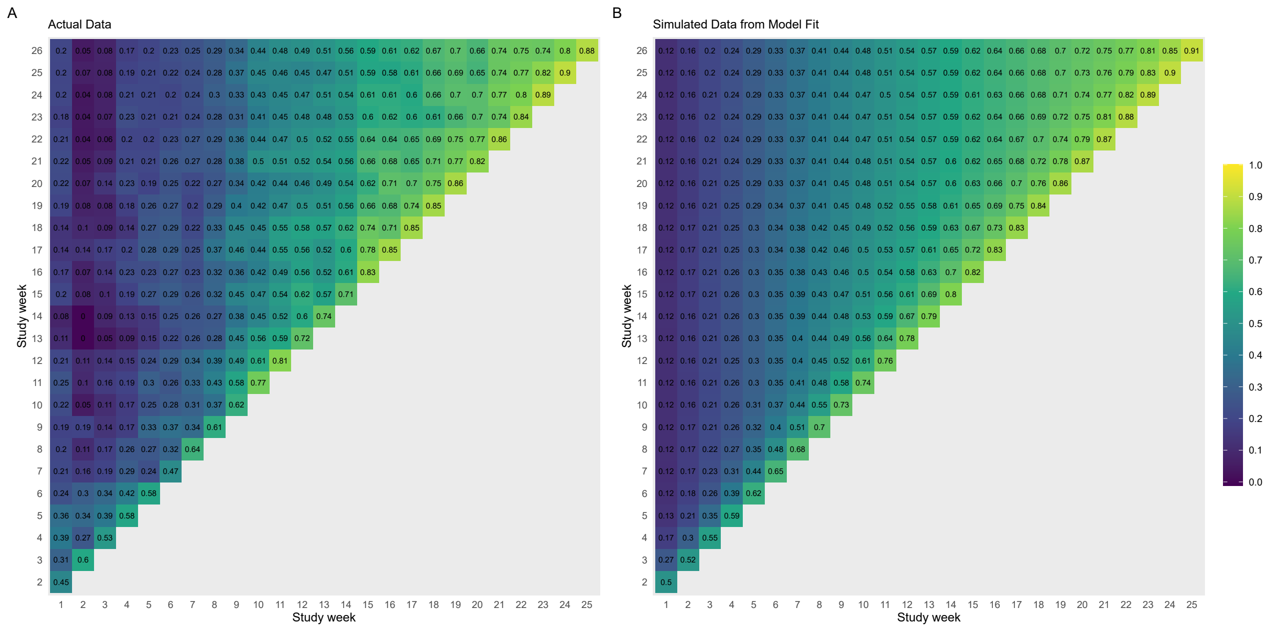


Figure 8: Panel (A) shows the Spearman correlation between outcomes on each par of study weeks using historical data. The correlation is high for nearby weeks, but quickly decays during the first 9 weeks. Panel (B) uses simulated data from the fitted model, resampling baseline covariates. We observe a decent match, indicating that our model is able to capture the correlation structure.

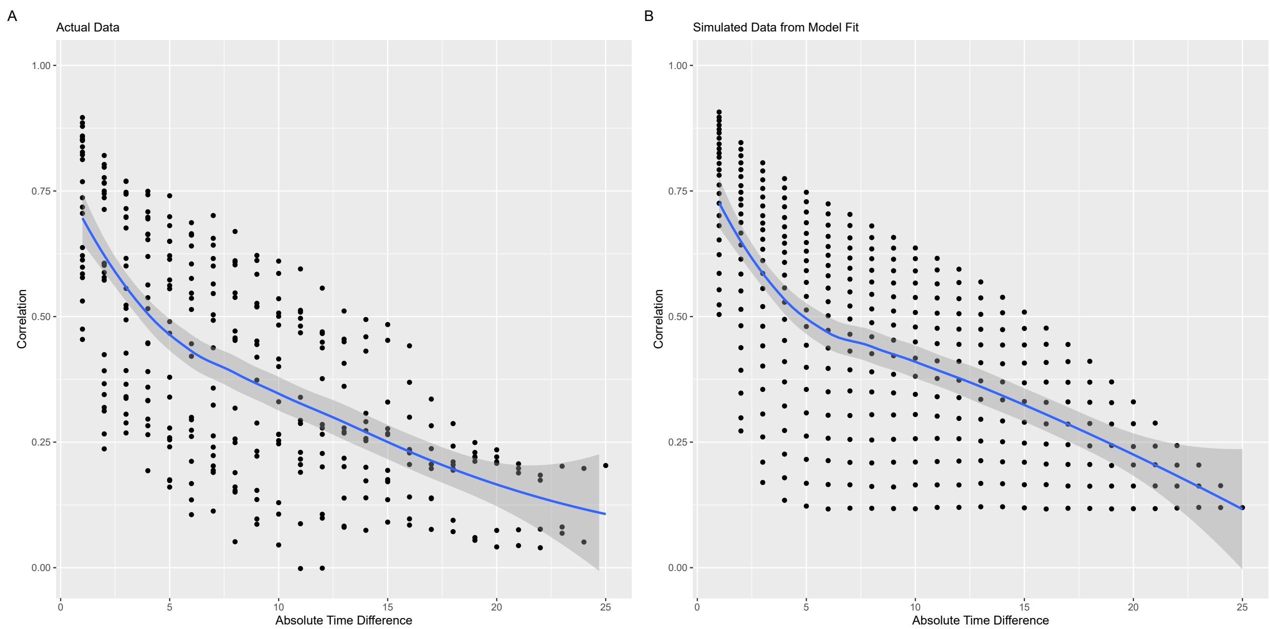


Figure 9: Comparison of variograms, which plot the mean Spearman correlation (y-axis) against the time lag in weeks between measurements (x-axis). Panel (A), using historical data, shows the empirical correlation

decaying as the time lag increases. Panel (B) shows the same plot for data simulated from the fitted model. The close alignment between the two plots suggests that the model effectively captures the decay in correlation over time.

7.3.3.7 Visualization of Estimate

We will visualize the SOPs and τ_{TAOOH} as shown in Figure 10 and Figure 11.

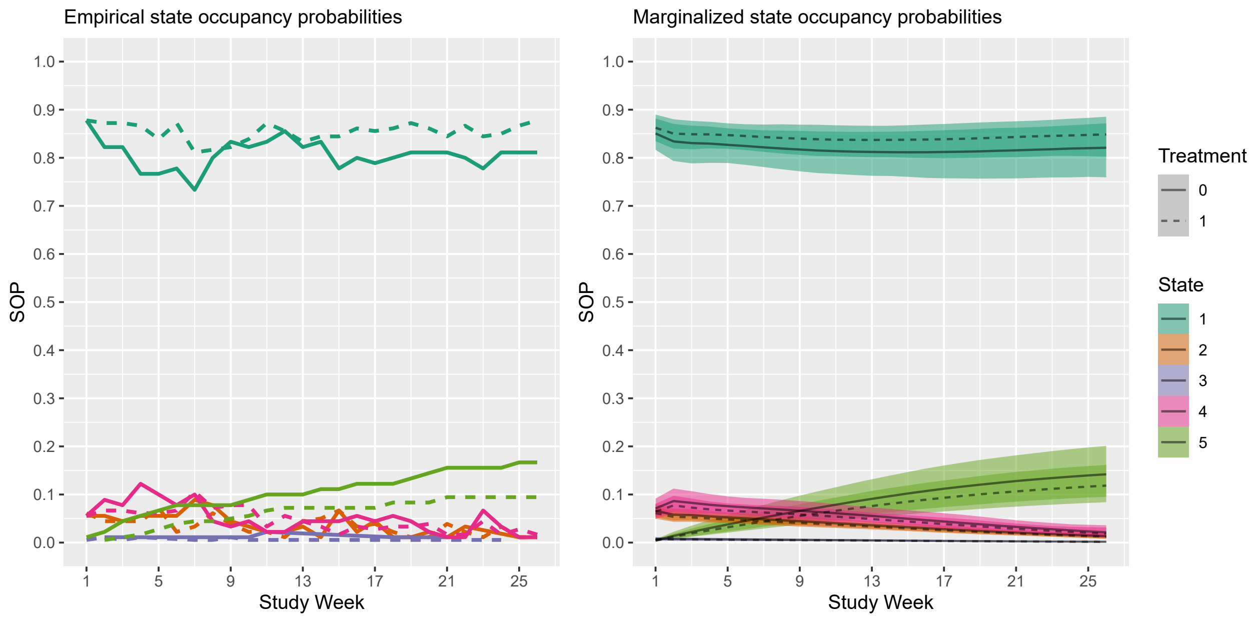


Figure 10: The left plot shows the empirical state occupancy probabilities (SOP) based on a sample ($n = 270$) from simulated data with a true underlying $OR = 0.8$ for transitioning to a worse state for the group selected for invitation to the single-arm trial ($\tau_{TAOOH} \approx 1.2$ weeks). The right plot shows the marginalized SOPs derived from the fitted model with 95 per cent credible bands.

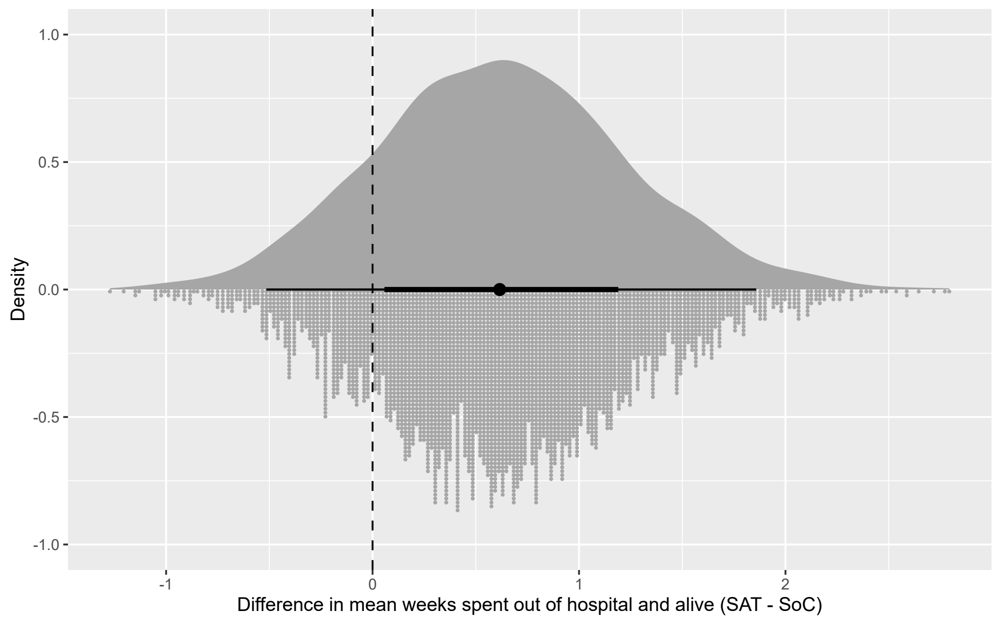


Figure 11: Posterior distribution of the primary estimand, the expected difference in weeks alive and out of hospital ($\hat{\tau}_{TAOOH}$), defined as state 1, based on a sample ($n = 270$) from simulated data with a true underlying $OR = 0.8$ for transitioning to a worse state for the group selected for invitation to the single-arm trial ($\tau \approx 1.2$ weeks). Larger values favour the group invited to the single-arm trial.

7.3.3.8 Operating Characteristics

Table 8 shows frequentist operating characteristics of the second-order transition model for TAOOH. To generate plausible trajectories, we fit a second-order markov model on historical data, and used its parameters to generated datasets with known treatment effects; $OR \in \{0.0, 0.7, 0.8, 0.9\}$. In our historical data, this approximately equaled the following differences in the weeks spent at home and alive over 26 weeks $\tau_{TAOOH} \in \{0, 0.6, 1.2, 1.8\}$. More information on the data generation can be found here. For each scenario, 1,000 datasets were simulated and analyzed using the second-order transition model (without covariates). The power is estimated as the proportion of simulations where the posterior probability of any benefit exceeds 0.95.

τ_{TAOOH}	OR	Assertion Probability (95% CI)
0 weeks	1.0	5.5% (4.1% - 6.9%)
~0.6 weeks	0.9	23.8% (21.3% - 26.3%)
~1.2 weeks	0.8	59.8% (56.7% - 62.9%)
~1.8 weeks	0.7	90.5% (88.7% - 92.3%)

Table 8: Probability of assertion of benefit when $P(\text{benefit}) > 0.95$ is chosen as a cut-off for.

We emphasize that the $P(\text{benefit}) > 0.95$ threshold is used here only for the purpose of evaluating operating characteristics. It will not be used as a formal, dichotomous decision rule in the final analysis. Instead, the primary reporting will focus on presenting and interpreting the full posterior distribution of $\hat{\tau}_{TAOOH}$. The estimates in Table 8 are intended to provide a conventional benchmark to illustrate which effect sizes the study can reasonably detect given our sample size and modeling approach.

8 Sensitivity Analyses

We will conduct sensitivity analyses for each of the three primary outcomes to assess the robustness of our findings to key modeling assumptions. These analyses are pre-specified but considered secondary to the primary analyses.

8.1 Overall Survival

As a sensitivity analysis for OS, we will fit a non-proportional hazards model to assess whether the proportional hazards assumption is violated. Specifically, we will model time-varying treatment effects using cubic B-splines with 3 degrees of freedom for the interaction between treatment and time.

8.2 Time Alive and Out of Hospital

As a sensitivity analysis for TAOOH, we will fit a first-order Markov model (conditioning only on y_{t-1} rather than both y_{t-1} and y_{t-2}). While the primary second-order model was selected based on model fit criteria in historical data, this sensitivity analysis will assess whether the additional complexity of the second-order specification materially affects our conclusions. If results are consistent between the first-order and second-order models, this would strengthen confidence in the robustness of our findings.

8.3 Quality of Life

We will conduct two sensitivity analyses for the Quality of Life endpoint to assess the robustness of our results to the Principal Stratum assumption and the handling of death.

8.3.1 Landmark Analysis

The first sensitivity analysis is a landmark analysis of Quality of Life up to 6 months. We define the outcome as the latest available Quality of Life measurement for each participant within the 6-month follow-up period, excluding the baseline measurement. To account for survival, we include death as the worst possible category

(State 8), ranked below “Very Poor” (State 7). We will compare the distribution of this composite outcome between the two groups using a proportional odds model.

8.3.2 Longitudinal Ordinal Model with Death

The second sensitivity analysis aims to fit a longitudinal ordinal model similar to the main analysis but incorporating death as an absorbing state. As the differential missingness between the absorbing state (death, which is always observed) and non-absorbing QoL states (which are sparsely observed) leads to biased estimates of the cumulative incidence of death if unobserved intervals are simply treated as censored (see Section 13.5), we will use multiple imputation. We will impute QoL values (states 1-7) for all weeks where patients are known to be alive but have no recorded measurement. This approach ensures that the cumulative incidence of death is correctly estimated. We will then fit the first-order Markov ordinal transition model to the imputed datasets, including death as state 8.

9 Secondary Outcomes

The following secondary outcomes are considered exploratory and focus on the implementation and feasibility of ctDNA-guided care, as well as downstream clinical processes. All analyses will be conducted within a Bayesian framework using default or weakly informative priors from `brms` (Bürkner 2017) and `rstanarm` (Goodrich et al. 2024). We will report posterior medians and 95% credible intervals.

9.1 Implementation Outcomes

9.1.1 Invitation Offered by Physician

Among patients selected for invitation, we estimate the proportion who were offered participation by their physician using a Bayesian multilevel logistic regression:

$$\begin{aligned}\text{Offered}_i &\sim \text{Bernoulli }(p_i)\\ \text{logit }(p_i) &= \beta_0 + \beta_1 \text{age}_i + \beta_2 \text{sex}_i\end{aligned}$$

where i indexes patients.

Derived Quantities: We will report the marginal probability of being offered participation. Exploratory analyses will assess differences by sex and age group.

9.1.2 Patient Acceptance of Invitation

Among patients who were offered participation, we estimate the proportion who provided informed consent using a Bayesian multilevel logistic regression:

$$\begin{aligned}\text{Accepted}_i &\sim \text{Bernoulli }(p_i)\\ \text{logit }(p_i) &= \beta_0 + \beta_1 \text{age}_i + \beta_2 \text{sex}_i\end{aligned}$$

where i indexes patients.

Derived Quantities: We will report the marginal probability of acceptance. Exploratory analyses will assess differences by sex and age.

9.1.3 Technical Feasibility of ctDNA Analysis

We assess three aspects of technical feasibility among SAT participants:

9.1.3.1 Proportion of Samples Without Error

$$\begin{aligned}\text{Error}_{ij} &\sim \text{Bernoulli }\left(p_{ij}\right)\\ \text{logit }\left(p_{ij}\right) &= \beta_0 + u_{0j}\\ u_{0j} &\sim \mathcal{N}(0, \sigma_u^2)\end{aligned}$$

where i indexes samples and j indexes patients, and u_{0j} is a patient-specific random intercept.

Derived Quantities: We will report the marginal probability of a technical error.

9.1.3.2 Proportion with ctDNA Detected at Baseline

$$\text{Detected}_i \sim \text{Bernoulli } (p_i)$$
$$\text{logit } (p_i) = \beta_0 + \mathbf{X}_i \boldsymbol{\beta_1}$$

where i indexes patients, $\mathbf{X}_i$ represents histological types of the a patient's cancer with coefficients $\boldsymbol{\beta_1}$.

Derived Quantities: We will report the marginal probability of ctDNA detection at baseline. Exploratory analyses will evaluate differences by histological subtype.

9.1.3.3 Proportion of Variants Suspected of CHIP

Using a multilevel logistic regression with random intercepts for patients and genes, adjusting for age:

$$\text{CHIP}_{ijk} \sim \text{Bernoulli } \left(p_{ijk}\right)$$
$$\text{logit } \left(p_{ijk}\right) = \beta_0 + f\left(\text{age}_j\right) + u_{0j} + v_{0k}$$
$$u_{0j} \sim \mathcal{N}\left(0, \sigma_u^2\right)$$
$$v_{0k} \sim \mathcal{N}\left(0, \sigma_v^2\right)$$

where i indexes variants, j indexes patients, k indexes genes, $f(\cdot)$ is a natural cubic spline with 3 knots for age, u_{0j} is a patient-level random intercept, and v_{0k} is a gene-level random intercept.

Derived Quantities: We will report the marginal probability of variants being suspected CHIP. Exploratory analyses assess whether CHIP prevalence increases with age. We will also examine the gene-level random effects to identify genes with substantially higher or lower CHIP probabilities.

9.1.4 Actionable Alterations

We estimate the proportion of SAT participants with at least one actionable alteration identified via ctDNA by cancer type:

$$\text{Actionable}_{ij} \sim \text{Bernoulli } \left(p_{ij}\right)$$
$$\text{logit } \left(p_{ij}\right) = \beta_0 + u_{0j}$$
$$u_{0j} \sim \mathcal{N}\left(0, \sigma_u^2\right)$$

where i indexes patients, j indexes cancer diagnosis categories, and u_{0j} is a diagnosis-specific random intercept.

Derived Quantities: We will report the marginal probability of having at least one actionable alteration, with exploratory analyses by cancer diagnosis category.

9.2 Comparative Outcomes (SAT vs. External Control)

9.2.1 Time to Treatment Switch or Stop

We evaluate the time from treatment initiation to the first occurrence of treatment switch or discontinuation. We estimate the cause-specific hazard, i.e., we censor patients who die prior to switching or stopping.

We use the same Bayesian proportional hazards model specified for Overall Survival (Section 7.1.3). The primary estimand is the cause-specific hazard ratio.

9.2.2 Time to Best Supportive Care

We evaluate the time from first-line treatment initiation to the switch to best supportive care. To specifically assess the timing of the decision to transition to palliative care (as a proxy for recognizing ineffective therapy) conditional on the patient being alive, we estimate the cause-specific hazard, i.e., patients who die without having switched to best supportive care are censored at the date of death.

We use the same Bayesian proportional hazards model as specified for Overall Survival (Section 7.1.3). We will report the posterior distribution of the cause-specific hazard ratio.

9.2.3 Targeted Therapy Use

We compare the probability of receiving targeted therapy between groups:

$$\begin{aligned} \text{Targeted}_i &\sim \text{Bernoulli}\ (p_i) \\ \text{logit}\ (p_i) &= \beta_0 + \beta_{\text{grp}} \cdot \text{grp}_i \end{aligned}$$

where i indexes patients. We do not add covariates to this model due to the small expected number of events.

Derived Quantities: We report the marginal probability of receiving targeted therapy in each group and the risk difference.

9.2.4 Blood Product Transfusions

We compare the rate of blood product transfusions (red blood cells or platelets) using a Bayesian negative binomial regression with an exposure offset for follow-up time.

$$\begin{aligned} Y_i &\sim \text{NegBinom}\ (\mu_i, \phi) \\ \log(\mu_i) &= \log(t_i) + \beta_0 + \beta_{\text{grp}} \cdot \text{grp}_i + f_1(\text{hgb}_i) + f_2(\text{plt}_i) + \\ &\quad f_3(\text{age}_i) + f_4(\text{crea}_i) + \sum_{k=2}^{3} \beta_{\text{ecog},k} \cdot \text{ecog}_{ik} \end{aligned}$$

where Y_i is the count of transfusions for patient i, t_i is the follow-up time, grp_i is the treatment group indicator, hgb_i is the hemoglobin concentration at baseline, plt_i is the platelet count at baseline, age_i is the age at baseline, $crea_i$ is creatinine at baseline, and ϕ is the overdispersion parameter. The functions $f_1(\cdot)$ to $f_4(\cdot)$ are modeled using restricted cubic splines with 3 knots.

Derived Quantities: The marginal rate ratio and difference in the rate of transfusions between groups will be reported.

10 Long-term Follow-up Analyses

We will repeat all analyses described above for follow-up periods of 12 months and 24 months. The estimands remain the same, extending the time horizon t_{max} to 365 days (12 months) and 730 days (24 months), respectively.

The statistical models will remain the same as specified for the primary 6-month analysis, with adjustments to the flexibility of the time-modeling components to account for the longer follow-up duration:

- **Overall Survival:** The degrees of freedom for the M-splines modeling the baseline hazard will be increased by one for the 12-month analysis (6 df) and by two for the 24-month analysis (7 df).
- **Quality of Life & Time Alive and Out of Hospital:** The restricted cubic splines modeling the effect of time in the transition models will use one additional knot for the 12-month analysis (5 knots) and two additional knots for the 24-month analysis (6 knots).

11 Limitations

11.1 QoL Analysis Limitations

11.1.1 Principal Stratum Assumption:

The primary QoL analysis relies on the strong, untestable assumption that selection for invitation has no effect on 6-month survival. If this assumption is violated, the analysis results apply only to the specific subgroup who would survive regardless, but this subgroup is not fully identifiable, and the estimate may be biased for the broader ITT population. Consequently, we will compare QoL using the principal stratum approach only if there is little evidence of a survival difference between groups at 6 months and pivot to the alternative analyses described above if the assumption seems implausible, given the data.

11.1.2 Informative Missingness

A primary limitation arises because the treatment strategy is expected to influence post-randomization clinical decisions, which in turn affect whether the QoL outcome is observed. These decisions affect both future QoL and the probability that QoL is measured, since data is collected during routine care appointments. For instance, a decision to switch a patient to best supportive care may lead to fewer clinic visits, resulting in the cessation of QoL data collection.

As a simple illustrative example, assume ctDNA guided management mainly improves QoL through an early switch to best supportive care of an ineffective treatment. Let's also assume that all the patients who are switched to best supportive care have no more follow-up visits, and their improved QoL is not recorded. Logically, this results in an inability to show the true treatment effect. Figure 12 shows this graphically: M_Y denotes the missingness indicator for the quality of life outcome Y. A represents the selection for invitation to the trial (treatment), and X represents measured baseline covariates. W is a vector of unmeasured causes of missingness (M_Y). The variable L represents post-randomization events (e.g., early switch to best supportive care) that are influenced by the treatment A and subsequently affect both the outcome Y and its missingness M_Y. The direct path $A \to Y$ captures effects of the intervention on quality of life that do not operate through mechanisms affecting missingness, although we anticipate such effects to be rare.

In LIQPLAT this problem will not be as severe, because the University Hospital does not have a separate unit for best supportive care. This means that many patients who are switched to best supportive care still have appointments and will be offered the QoL questionnaire.

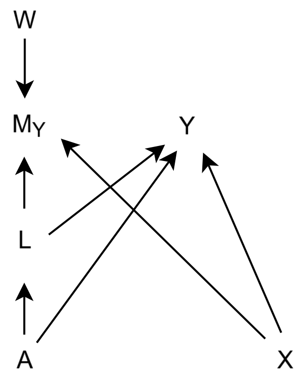


Figure 12: A missingness Directed Acyclic Graph (m-DAG) depicting assumptions regarding missingness in the quality of life outcome (Y). M_Y is the missingness indicator for Y. W is the vector of unmeasured causes of M_Y. A represents the selection for invitation to LIQPLAT (treatment indicator), X represents measured baseline covariates, L represents effects of A which affect both follow-up and thus missingness (M_Y) and QoL (Y), such as early switch to best supportive care. $A \to Y$ represent effects of A on Y which do not affect missingness, though we expect these to be rare. Realistically, ctDNA sampling would mostly affect Y through changes in the treatment regimen, which also affects missingness.

12 Reporting

Results will be reported according to CONSORT guidelines where applicable, adapted to the SAT design. Posterior distributions for primary estimands will be visualized (e.g., density plots) and summarized (median, 95% CrI).

13 Appendices

13.1 Alternative Estimands

While the primary analysis defines the treatment as the random selection for invitation, an alternative estimand of clinical interest is the average causal effect of the intervention among patients who actually accepted the invitation.

We consider an extended causal diagram, where Z denotes random selection for invitation to the SAT, I denotes invitation to the SAT, A denotes acceptance of said invitation, and Y denotes the coutome. S is an indicator for selection into the population which accepted the invitation. L_1 and L_2 are measured (baseline) covariates which capture the effects of the unmeasured causes U on A and I. By adjusting for L_1 and L_2, we could block the paths $Z \rightarrow \boxed{S} \leftarrow I \leftarrow L_1 \leftarrow U \rightarrow Y$ and $Z \rightarrow \boxed{S} \leftarrow A \leftarrow L_2 \leftarrow U \rightarrow Y$. However, we have not evaluated / assessed whether our routine data adequately captures L_1 and L_2 and therefore do not plan to estimate this estimand.

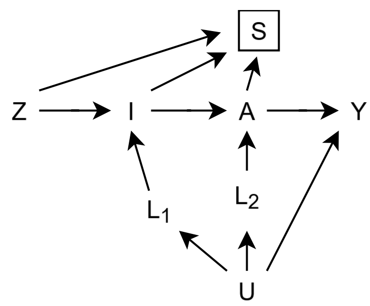


Figure 13: Extended causal diagram for alternative estimands. Z denotes random selection, I denotes actual invitation, and A denotes acceptance. U represents unmeasured common causes of the treatment process and the outcome Y. L_1 and L_2 are measured covariates that act as mediators for U; L_1 predicts the actual invitation, while L_2 predicts acceptance. S represents the selection into the analyzed population.

13.2 Multiple Imputation of Quality of Life Data

13.2.1 Overview

This appendix describes the multiple imputation strategy for the Quality of Life (QoL) outcome in the LIQPLAT trial. We use Multiple Imputation by Chained Equations (MICE) with multilevel predictive mean matching (PMM) to handle missing data under the Missing At Random (MAR) assumption.

13.2.2 Data

The following code uses historical data from the Division of Medical Oncology at USB (first appointments from 2023 to mid-2024, prior to LIQPLAT initiation). These data are used to develop and validate the imputation procedure.

13.2.3 Data Preprocessing

13.2.3.1 Inclusion Criteria

The substantive analysis model includes only patients who survived beyond 182 days. To maintain congruence between the imputation and analysis models, we restrict the imputation dataset to:

1. Observations up to day 182 after random selection
2. Patients who either have no recorded death or died after day 182

13.2.3.2 Variable Transformations

Ordered factors (ECOG, QoL, distress) are converted to numeric format for imputation. This avoids sparse dummy variable matrices that can cause convergence issues in MICE. Predictive mean matching preserves the discrete nature of these variables by imputing only observed values.

```
imputation_vars <- c(
  "pat_id",
  "tx",
  "gender",
  "age",
  "ecog_fstcnt",
  "diagnosis",
  "plan_fstcnt_coded",
  "week",
  "distress",
  "q30"
  # Variables available in trial data but not in historical dataset:
  # "albumin",
  # "c_reactive_protein"
)

df <- df_raw |>
  filter(quest_day <= 182, (is.na(death_day) | death_day >= 182)) |>
  mutate(
    tx = if_else(pat_id %in% tx, "1", "0"),
    distress = as.numeric(as.factor(coalesce(distress_therm_2, distress_therm_v2,
distress_therm_inpat)), ordered = TRUE),
    week = ceiling(quest_day / 7),
    diagnosis = fct_lump_n(diagnosis, n = 4),
    diagnosis = factor(as.integer(diagnosis)),
    plan_fstcnt_coded = fct_lump_n(plan_fstcnt_coded, n = 4),
    plan_fstcnt_coded = factor(as.integer(plan_fstcnt_coded)),
    q30       = as.numeric(as.character(q30)),
```

```r
    ecog_fstcnt = as.numeric(as.character(ecog_fstcnt)),
    tx = factor(tx)
    ) |>
  rename(age = pat_age) |>
  select(all_of(imputation_vars)) |>
  arrange(pat_id, week)

# Extract baseline QoL: last observation at or before random selection (week <= 0)
baseline_qol <- df |>
  filter(week <= 0) |>
  select(pat_id, q30, week) |>
  group_by(pat_id) |>
  arrange(desc(week)) |>
  slice_head(n = 1) |>
  ungroup() |>
  select(pat_id, baseline_q30 = q30)

df <- df |>
  left_join(baseline_qol, by = "pat_id") |>
  filter(week > 0) |>
  relocate(baseline_q30, ecog_fstcnt, diagnosis, plan_fstcnt_coded, .after = pat_id)

# Model non-linear time effects using restricted cubic splines with 4 knots
basis_rcs <- rms::rcs(df$week, 4)

df$week <- as.vector(basis_rcs[, 1])
df$week_nl1 <- as.vector(basis_rcs[, 2])
df$week_nl2 <- as.vector(basis_rcs[, 3])
```

13.2.4 Imputation Variables

The imputation model includes variables from the substantive analysis model and auxiliary variables which either causally affect the missing values or their missingness indicator (Mathur, Zhang, and Shpitser 2025; Buuren 2021).

13.2.4.1 Analysis Variables

Variable	Description	Level
`tx`	Treatment group allocation	Baseline
`age`	Age at date of random selection (years)	Baseline
`gender`	Patient sex	Baseline
`ecog_fstcnt`	ECOG performance status at baseline (0–4)	Baseline
`diagnosis`	Primary cancer diagnosis (5 categories)	Baseline
`baseline_q30`	EORTC QLQ-C30 Q30 at baseline	Baseline
`week`	Week of questionnaire completion (post-random selection)	Longitudinal
`q30`	EORTC QLQ-C30 question 30 (overall QoL, 1–7)	Longitudinal

Table 9: Analysis variables included in the imputation model

13.2.4.2 Auxiliary Variables

Variable	Description	Level
`distress`	Distress thermometer score (0–10)	Longitudinal
`plan_fstcnt_coded`	Treatment intent at baseline (5 categories)	Baseline
`albumin`	Serum albumin at baseline (g/L)	Baseline
`c_reactive_protein`	C-reactive protein at baseline (mg/L)	Baseline

Table 10: Auxiliary variables included in the imputation model. Albumin and CRP are available in trial data but not in the historical dataset used here.

13.2.5 Multiple Imputation Procedure

We use multilevel multiple imputation to account for the hierarchical data structure, where repeated QoL measurements (Level 1) are nested within patients (Level 2). The patient identifier (`pat_id`) serves as the clustering variable.

13.2.5.1 Imputation Methods

Baseline variables (Level 2) are imputed using `2lonly.pmm` from the `miceadds` package. This method:

1. Aggregates longitudinal predictors to the cluster level by computing patient means
2. Fits a linear regression model on the collapsed data:

$$Y_j = \beta_0 + \mathbf{X}_j \boldsymbol{\beta_1} + \bar{\mathbf{Z}}_j \boldsymbol{\beta_2} + \epsilon_j, \quad \epsilon_j \sim N(0, \sigma_\epsilon^2)$$

where Y_j is the missing baseline variable for patient j, $\mathbf{X}_j$ represents other baseline predictors, and $\bar{\mathbf{Z}}_j$ represents aggregated means of longitudinal predictors

3. Selects imputed values via predictive mean matching

Longitudinal variables (Level 1) are imputed using `2l.pmm`, which fits a linear mixed-effects model:

$$Y_{ij} = \beta_0 + \mathbf{X}_{ij} \boldsymbol{\beta_1} + u_{0j} + \epsilon_{ij}, \quad u_{0j} \sim N(0, \sigma_u^2), \quad \epsilon_{ij} \sim N(0, \sigma_\epsilon^2)$$

where Y_{ij} is the outcome for observation i of patient j, $\mathbf{X}_{ij}$ includes fixed effects (time-varying and static predictors), and $u_{0j} \sim N(0, \sigma_u^2)$ is the patient-specific random intercept.

For both methods, we use Type 1 predictive mean matching with a donor pool of $k = 5$ (Buuren 2021, chapters 3.4, 7.8).

13.2.5.2 Predictor Matrix Configuration

The predictor matrix is configured to:

- Use `pat_id` as the clustering variable (coded as -2)
- Prevent week and it's transformations from predicting static baseline variables
- Allow baseline variables to predict longitudinal outcomes

```
#| Note: q30 is already inverted so that higher values indicate lower QoL
md.pattern(df)
```

```
init <- mice(df, maxit = 0)
pred_matrix <- init$predictorMatrix
meth <- init$method
```

```r
# Set patient ID as clustering variable
pred_matrix[, "pat_id"] <- -2
pred_matrix["pat_id", ] <- 0

# Time variables should not predict static baseline variables
baseline_vars <- c("tx", "age", "gender", "diagnosis", "ecog_fstcnt",
"plan_fstcnt_coded", "baseline_q30")
pred_matrix[c(baseline_vars, "week", "week_nl1", "week_nl2"), c("week", "week_nl1",
"week_nl2")] <- 0

# Longitudinal variables: multilevel PMM
meth["q30"] <- "2l.pmm"
meth["distress"] <- "2l.pmm"

# Baseline variables: cluster-level PMM
meth["ecog_fstcnt"] <- "2lonly.pmm"
meth["plan_fstcnt_coded"] <- "2lonly.pmm"
meth["diagnosis"] <- "2lonly.pmm"
meth["baseline_q30"] <- "2lonly.pmm"


imputed_data <- mice(df,
                     m = 5,         # 50 in final analysis
                     maxit = 5,     # 50 in final analysis
                     predictorMatrix = pred_matrix,
                     method = meth,
                     donors = 5,
                     seed = 78109823)
```

13.2.5.3 Imputation Parameters

For the final trial analysis:

- Number of imputed datasets: $m = 50$
- Maximum iterations: 50
- Donor pool size: $k = 5$

13.2.6 Convergence Diagnostics

Convergence is assessed via trace plots of the mean and standard deviation of imputed values across iterations.

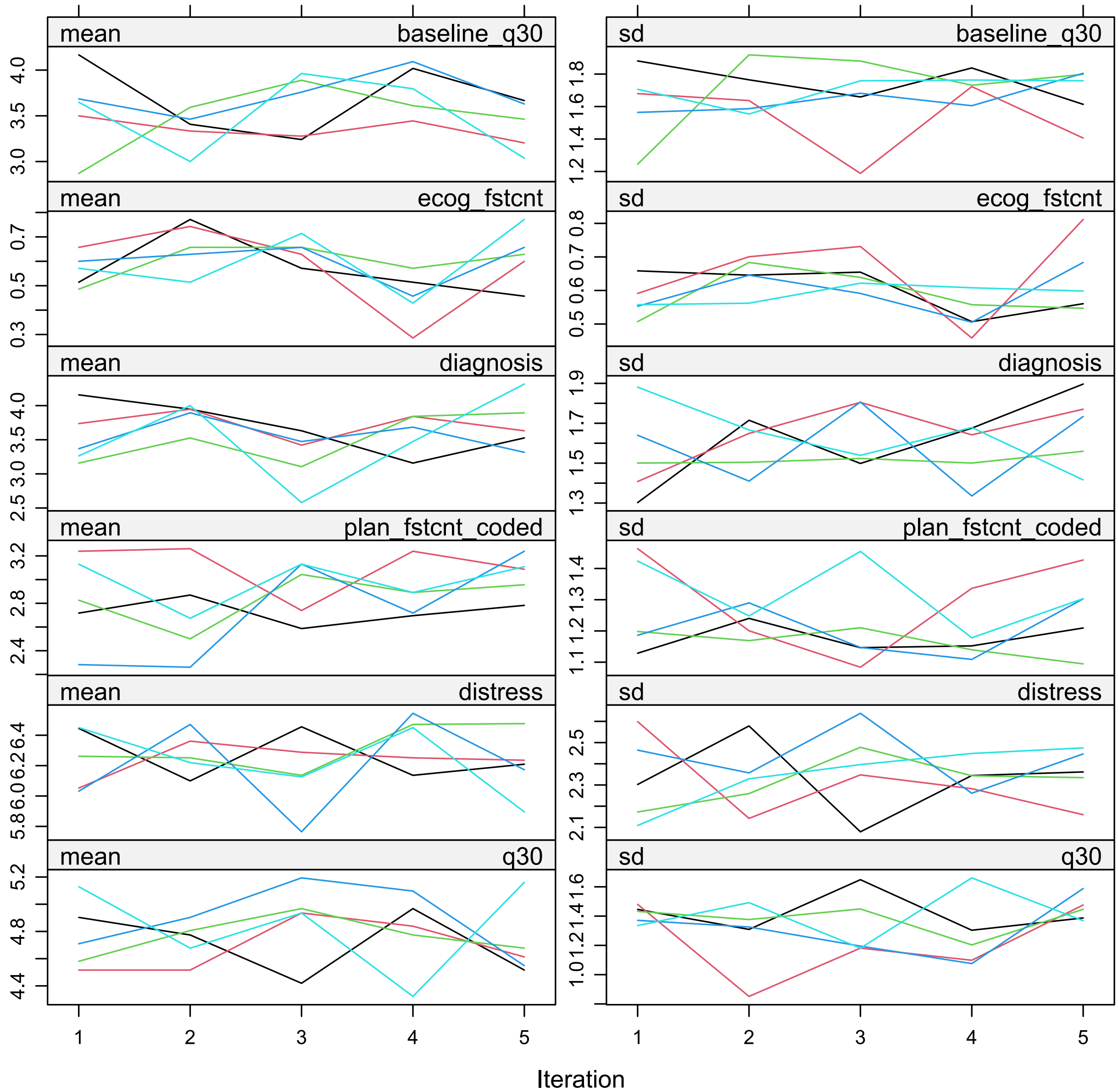


Figure 14: Trace plots showing the mean and standard deviation of imputed values across iterations for each imputed dataset.

13.2.7 Distribution Diagnostics

We compare the distributions of observed and imputed values to assess plausibility.

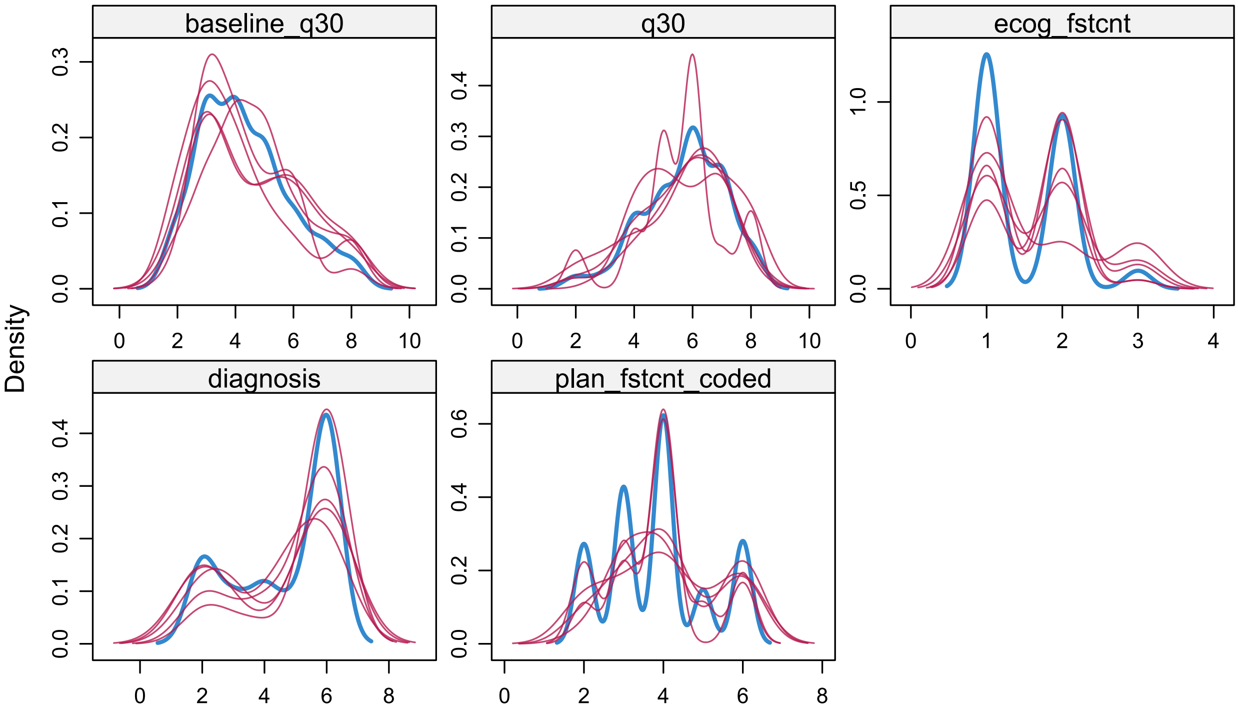


Figure 15: Density plots comparing observed (blue) and imputed (red) values for baseline variables and QoL.

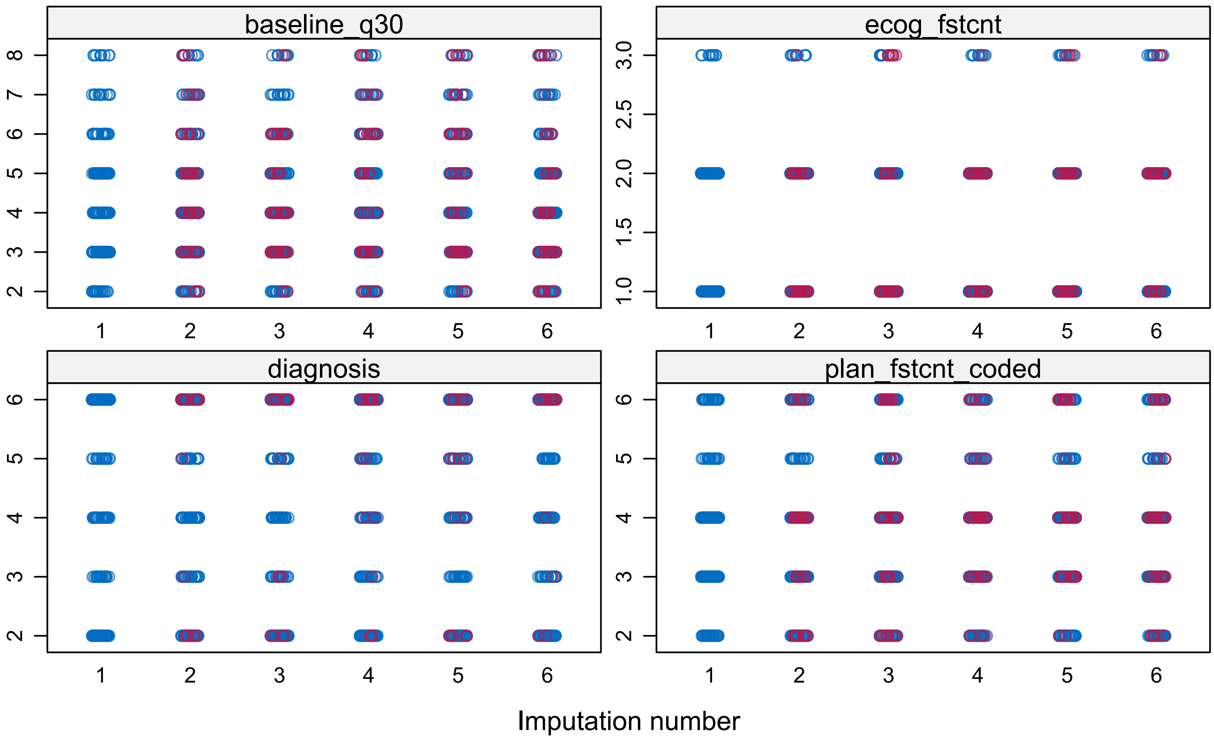


Figure 16: Strip plots of imputed values for baseline variables across imputed datasets.

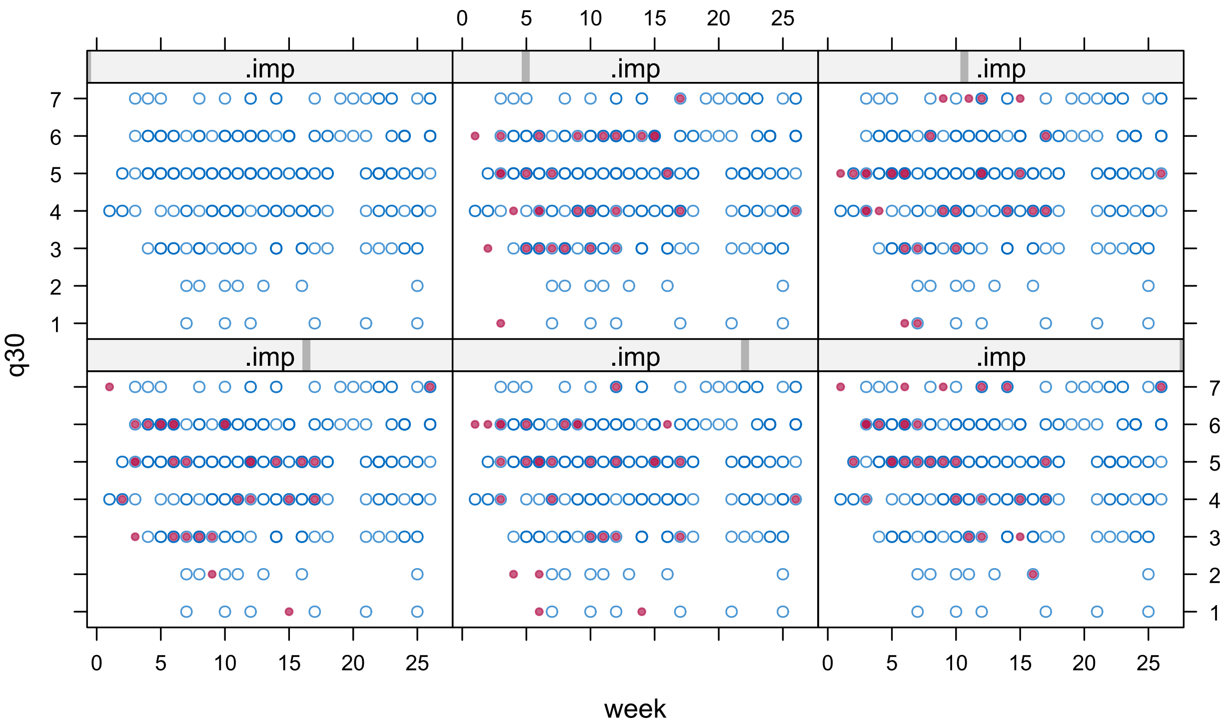


Figure 17: Scatter plots of QoL against time, by imputed dataset and treatment group, and against baseline QoL.

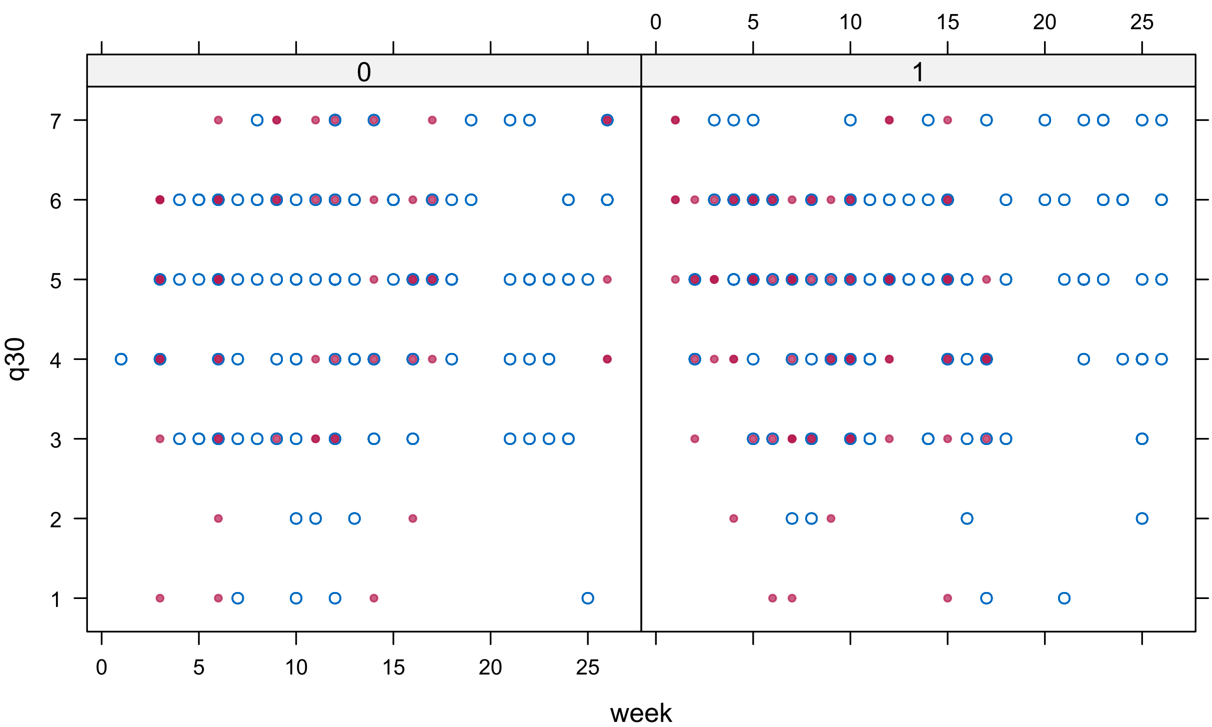


Figure 18: Scatter plots of QoL against time, by imputed dataset and treatment group, and against baseline QoL.

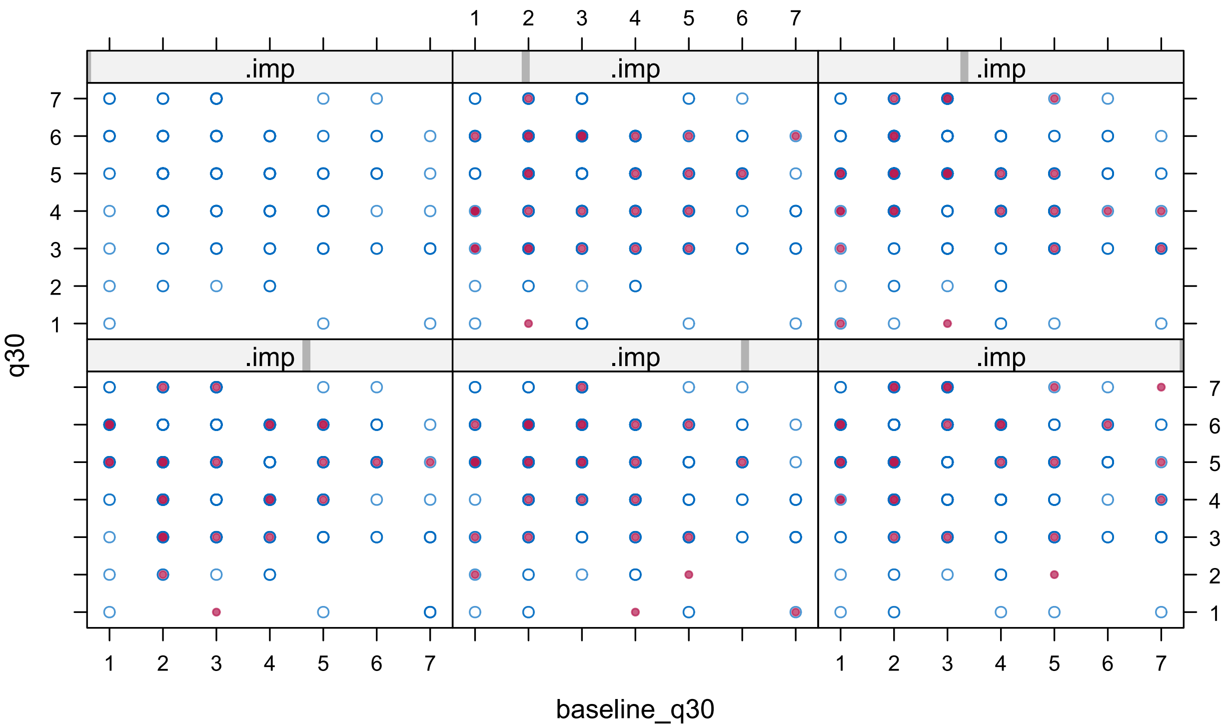


Figure 19: Scatter plots of QoL against time, by imputed dataset and treatment group, and against baseline QoL.

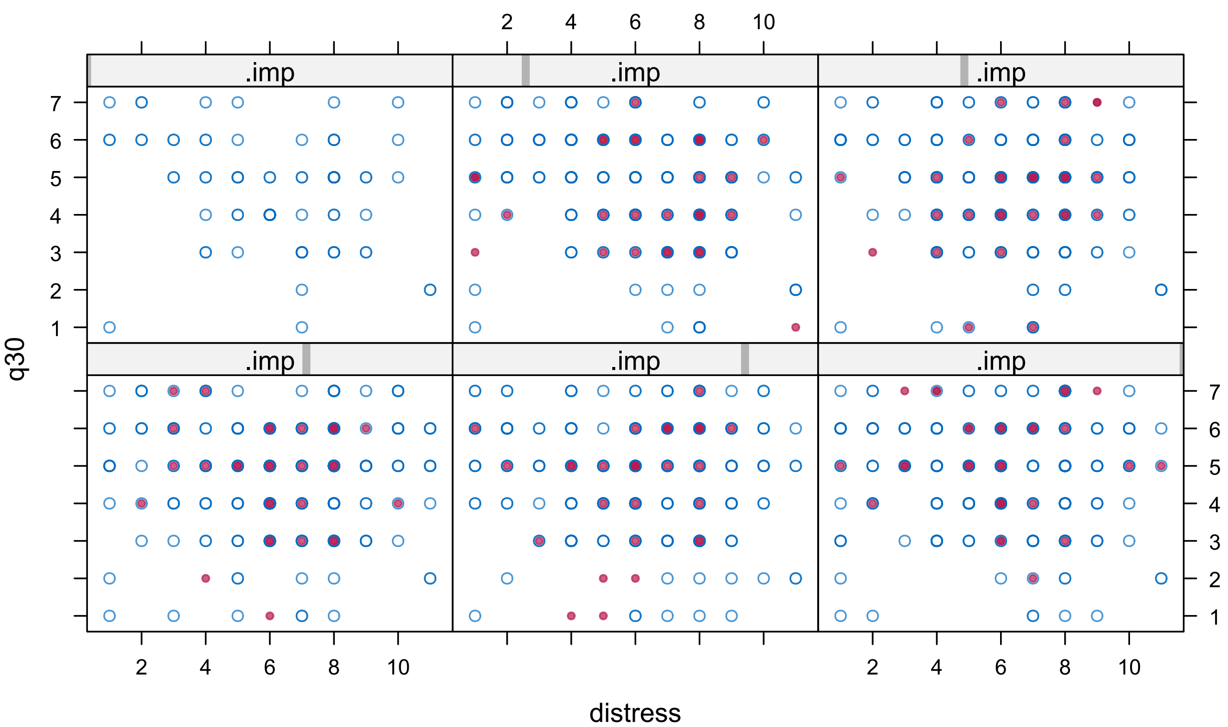


Figure 20: Scatter plot of QoL against distress thermometer score by imputed dataset.

13.2.8 Analysis of Trial Data

For the final trial analysis, all $m = 50$ imputed datasets will be analysed, and posterior draws will be stacked to propagate imputation uncertainty into the final inference.

13.3 Multiple Imputation Strategy for Time Alive and Out of Hospital

13.3.1 Overview

This appendix describes the multiple imputation strategy for the Time Alive and Out of Hospital (TAOOH) outcome in the LIQPLAT trial. The approach uses Multiple Imputation by Chained Equations (MICE) with multilevel predictive mean matching (PMM) to handle missing data under the Missing At Random (MAR) assumption.

The TAOOH outcome is a 5-level ordinal score constructed from the imputed binary indicators: (1) Alive and Out of Hospital, (2) Alive and In Hospital (Planned Admission), (3) Alive and In Hospital (ER Visit), (4) Alive and In Hospital (Unplanned Admission), (5) Dead. This ordinal score is derived *after* imputation of the component indicators.

13.3.2 Data

The following code uses historical data from the Division of Medical Oncology at USB (first appointments from 2023 to mid-2024, prior to LIQPLAT initiation). These data are used to develop and validate the imputation procedure.

13.3.3 Data Preprocessing

13.3.3.1 Variable Selection and Transformations

```
imputation_vars <- c(
  "pat_id",
  "tx",
  "gender",
  "ecog_fstcnt",
  "diagnosis",
  "age",
  "albumin",
  "bilirubin",
  "c_reactive_protein",
  "lactate_dehydrogenase",
  "day",
  "in_hospital",
  "unplanned_stay",
  "er_visit",
  "death"
)

df <- df_raw |>
  select(all_of(imputation_vars)) |>
  mutate(
    diagnosis = fct_lump_n(diagnosis, n = 4),
    diagnosis = factor(as.integer(diagnosis)),
    ecog_fstcnt = factor(ecog_fstcnt, levels = 1:3, ordered = TRUE), #In LIQPLAT there
might be patients with level 4
    in_hospital = case_when(
        in_hospital == 1 ~ TRUE,
        TRUE ~ FALSE),
    unplanned_stay = case_when(
        unplanned_stay == 1 ~ TRUE,
        TRUE ~ FALSE),
    er_visit = case_when(
        er_visit == 1 ~ TRUE,
```

```
        TRUE ~ FALSE),
      lactate_dehydrogenase = log(lactate_dehydrogenase + 1),
      tx = factor(tx)
      ) |>
    filter(day < 183) |>
    group_by(pat_id) |>
    mutate(
      survival_time = max(day),
      died = if_else(any(death == 1), 1, 0),
      week = ceiling(day / 7),
      survival_time = ceiling(survival_time / 7)
    ) |>
    group_by(pat_id, week) |>
    mutate(
      death = factor(max(death)),
      in_hospital = factor(max(in_hospital)),
      unplanned_stay = factor(max(unplanned_stay)),
      er_visit = factor(max(er_visit))
    ) |>
    ungroup() |>
    filter(week > -2) |>
    select(-day) |>
    distinct() |>
    arrange(pat_id, week)

# Model non-linear time effects using restricted cubic splines with 4 knots
basis_rcs <- rms::rcs(df$week, 4)

df$week <- as.vector(basis_rcs[, 1])
df$week_nl1 <- as.vector(basis_rcs[, 2])
df$week_nl2 <- as.vector(basis_rcs[, 3])
```

13.3.3.2 Nelson-Aalen Cumulative Hazard Estimate

We include the Nelson-Aalen cumulative hazard estimate (`na_est`) as a predictor for baseline variable imputation rather than using the `death` indicator directly. When imputing baseline variables using `2lonly.pmm`, longitudinal predictors are aggregated to patient-level means. A death indicator would thus turn into the inverse of the survival time, conditional of the event occuring, creating a very skewed predictor. We thus use the Nelson-Aalen estimate, which is a smoother representation of the cumulative hazard over time.

```
na_df <- df |>
    select(pat_id, survival_time, died) |>
    unique()

na_est <- cbind(na_df$pat_id, nelsonaalen(na_df, timevar = survival_time, statusvar =
died)) |>
    data.table() |>
    rename(pat_id = V1, na_est = V2)

df <- df |>
    left_join(na_est, by = "pat_id")
```

13.3.3.3 Simulating Missing Data for Validation

For validation purposes, we introduce 10% missing data in the longitudinal outcome variables. In actual trial data, missingness in these variables is expected to be minimal.

13.3.4 Imputation Variables

Following Mathur, Zhang, and Shpitser (2025), we restrict the imputation model to variables included in the analysis model or those acting causally on the incomplete variables or their missingness indicators.

13.3.4.1 Analysis Variables

Variable	Description	Level
`tx`	Treatment group allocation	Baseline
`ecog_fstcnt`	ECOG performance status at baseline (0–4)	Baseline
`diagnosis`	Primary cancer diagnosis (5 categories)	Baseline
`albumin`	Serum albumin at baseline (g/L)	Baseline
`c_reactive_protein`	C-reactive protein at baseline (mg/L)	Baseline
`week`	Week after randomisation (RCS with 4 knots)	Longitudinal
`in_hospital`	Indicator for hospitalisation	Longitudinal
`unplanned_stay`	Indicator for unplanned hospital admission	Longitudinal
`er_visit`	Indicator for emergency room visit	Longitudinal
`death`	Death indicator	Longitudinal

Table 11: Analysis variables included in the imputation model

13.3.4.2 Auxiliary Variables

Variable	Description	Level
`age`	Age at date of randomisation (years)	Baseline
`gender`	Patient sex	Baseline
`bilirubin`	Serum bilirubin at baseline (μmol/L)	Baseline
`lactate_dehydrogenase`	Lactate dehydrogenase at baseline (U/L, log-transformed)	Baseline
`na_est`	Nelson-Aalen cumulative hazard estimate	Post-Baseline

Table 12: Auxiliary variables included in the imputation model

13.3.5 Multiple Imputation Procedure

We use multilevel multiple imputation to account for the hierarchical data structure, where repeated weekly observations (Level 1) are nested within patients (Level 2). The patient identifier (`pat_id`) serves as the clustering variable.

13.3.5.1 Imputation Methods

Baseline variables (Level 2) are imputed using `2lonly.pmm` from the `miceadds` package. This method:

1. Aggregates longitudinal predictors to the cluster level by computing patient means
2. Fits a linear regression model on the collapsed data:

$$Y_j = \beta_0 + \mathbf{X}_j\boldsymbol{\beta_1} + \bar{\mathbf{Z}}_j\boldsymbol{\beta_2} + \epsilon_j, \quad \epsilon_j \sim N(0, \sigma_\epsilon^2)$$

where Y_j is the missing baseline variable for patient j, $\mathbf{X}_j$ represents other baseline predictors, and $\bar{\mathbf{Z}}_j$ represents aggregated means of longitudinal predictors

3. Selects imputed values via predictive mean matching

Longitudinal variables (Level 1) are imputed using `2l.pmm`, which fits a linear mixed-effects model:

$$Y_{ij} = \beta_0 + \mathbf{X}_{ij}\boldsymbol{\beta}_1 + u_{0j} + \epsilon_{ij}, \quad u_{0j} \sim N(0, \sigma_u^2), \quad \epsilon_{ij} \sim N(0, \sigma_\epsilon^2)$$

where Y_{ij} is the outcome for observation i of patient j, $\mathbf{X}_{ij}$ includes fixed effects (time-varying and static predictors), and $u_{0j} \sim N(0, \sigma_u^2)$ is the patient-specific random intercept.

For both methods, we use Type 1 predictive mean matching with a donor pool of $k = 5$ (Buuren 2021, chapters 3.4, 7.8).

13.3.5.2 Predictor Matrix Configuration

The predictor matrix is configured to:

- Use `pat_id` as the clustering variable (coded as -2)
- Exclude death-related variables from predictions (we use `na_est` instead)
- Prevent time variables from predicting static baseline variables

```
md.pattern(df)
```

```
init <- mice(df, maxit = 0)
pred_matrix <- init$predictorMatrix
meth <- init$method

# Use patient ID as clustering variable
pred_matrix[, "pat_id"] <- -2
pred_matrix["pat_id", ] <- 0

# Exclude raw death/survival variables from predictions (we use na_est instead)
pred_matrix[, "death"] <- 0
pred_matrix["death", ] <- 0
pred_matrix[, "died"] <- 0
pred_matrix["died", ] <- 0
pred_matrix[, "survival_time"] <- 0
pred_matrix["survival_time", ] <- 0

# week shouldn't predict static baseline variables nor itself
baseline_vars <- c("tx", "age", "gender", "diagnosis", "ecog_fstcnt", "albumin",
"bilirubin",
                   "c_reactive_protein", "lactate_dehydrogenase", "na_est")

pred_matrix[c(baseline_vars, "week", "week_nl1", "week_nl2"), c("week", "week_nl1",
"week_nl2")] <- 0

# treatment, gender, age, and week are never missing
meth["tx"] <- ""
meth["gender"] <- ""
meth["age"] <- ""
meth["week"] <- ""
meth["week_nl1"] <- ""
meth["week_nl2"] <- ""
```

```r
# use mean matching at the cluster level for baseline observations
meth["diagnosis"] <- "2lonly.pmm"
meth["ecog_fstcnt"] <- "2lonly.pmm"
meth["albumin"] <- "2lonly.pmm"
meth["bilirubin"] <- "2lonly.pmm"
meth["c_reactive_protein"] <- "2lonly.pmm"
meth["lactate_dehydrogenase"] <- "2lonly.pmm"

# Longitudinal variables: multilevel PMM
meth["in_hospital"] <- "2l.pmm"
meth["unplanned_stay"] <- "2l.pmm"
meth["er_visit"] <- "2l.pmm"


imputed_data <- mice(df,
                     m = 5,       # 50 in final analysis
                     maxit = 5,   # 50 in final analysis
                     predictorMatrix = pred_matrix,
                     method = meth,
                     donors = 5,
                     seed = 1234)
```

13.3.5.3 Imputation Parameters

For the final trial analysis:

- Number of imputed datasets: $m = 50$
- Maximum iterations: 50
- Donor pool size: $k = 5$

13.3.6 Convergence Diagnostics

Convergence is assessed via trace plots of the mean and standard deviation of imputed values across iterations.

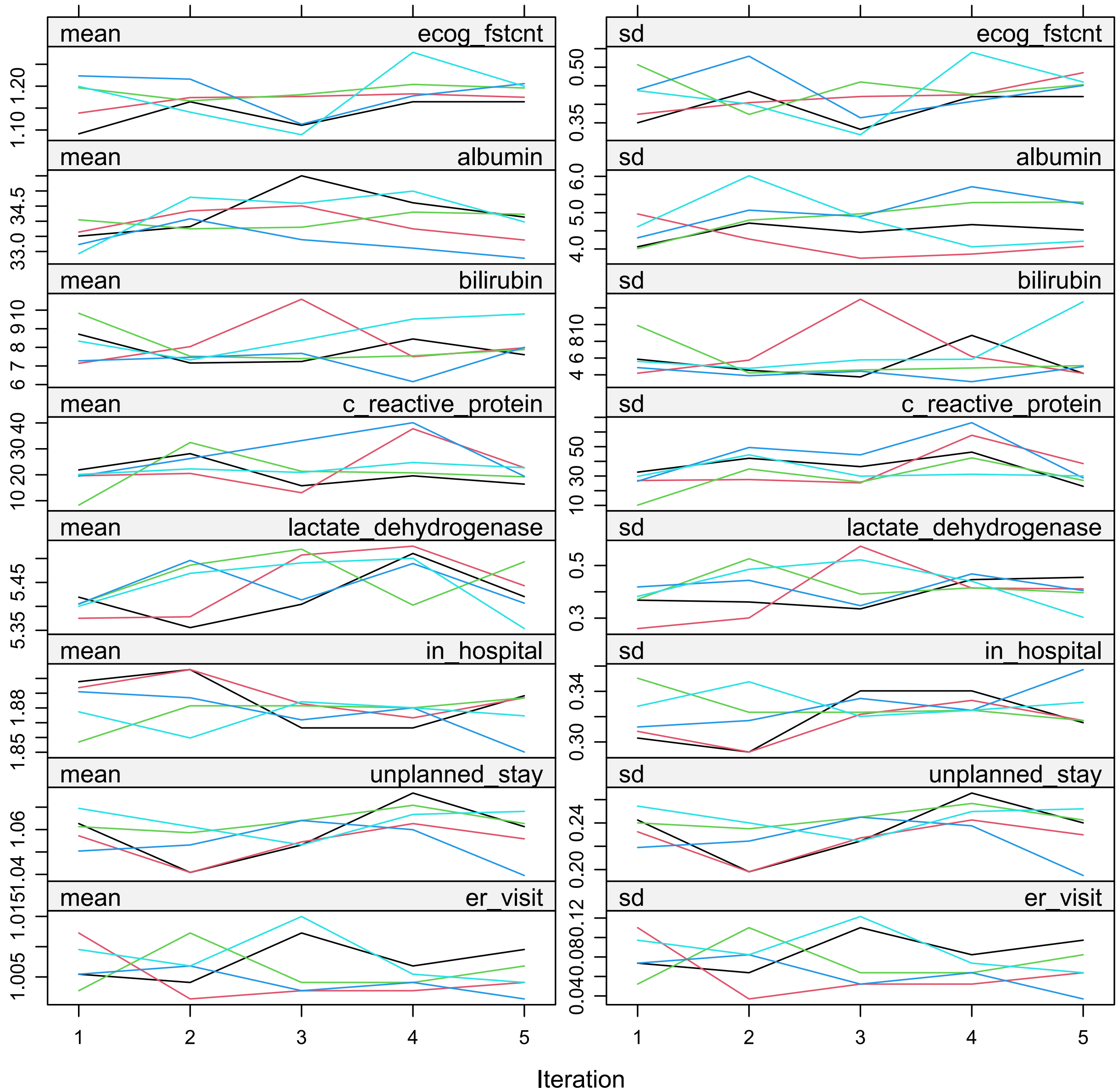


Figure 21: Trace plots showing the mean and standard deviation of imputed values across iterations for each imputed dataset.

13.3.7 Distribution Diagnostics

We compare the distributions of observed and imputed values to assess plausibility.

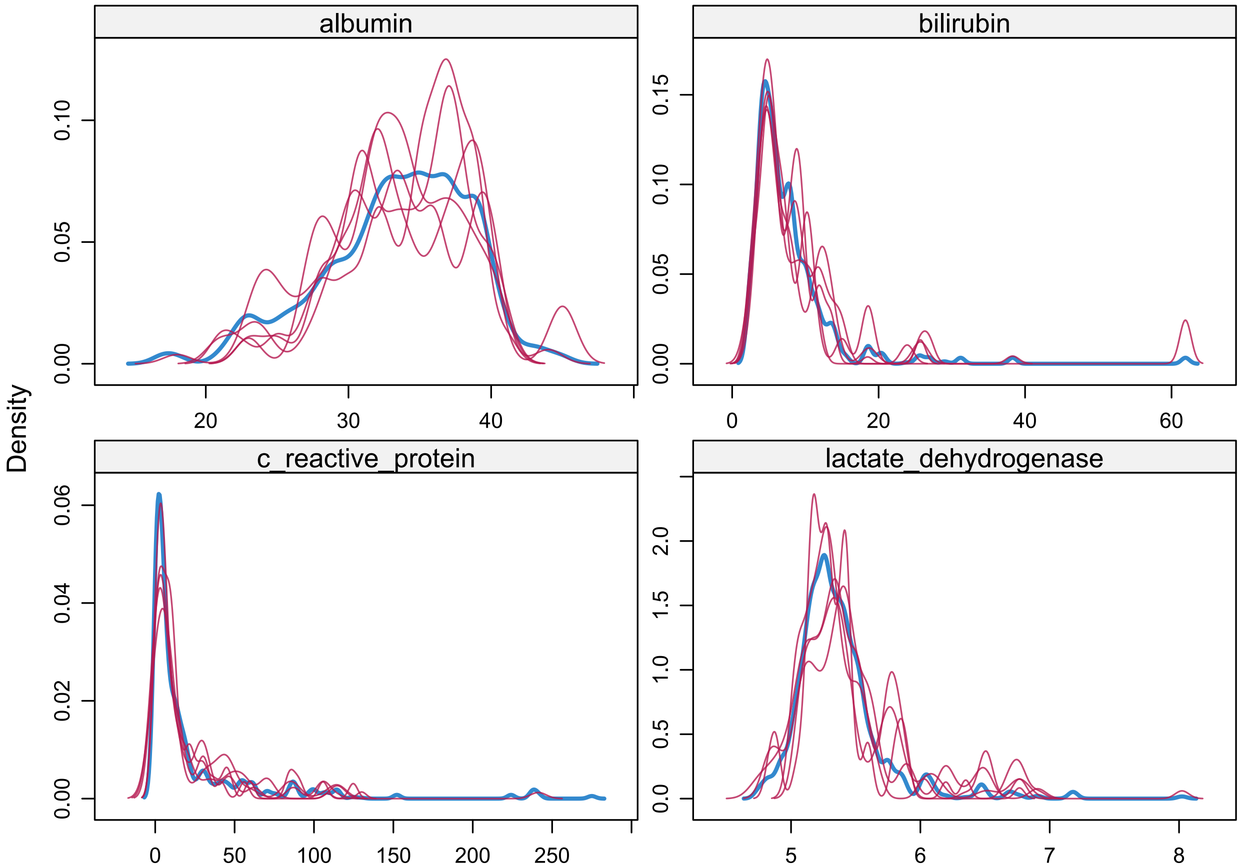


Figure 22: Density plots comparing observed (blue) and imputed (red) values for all imputed variables.

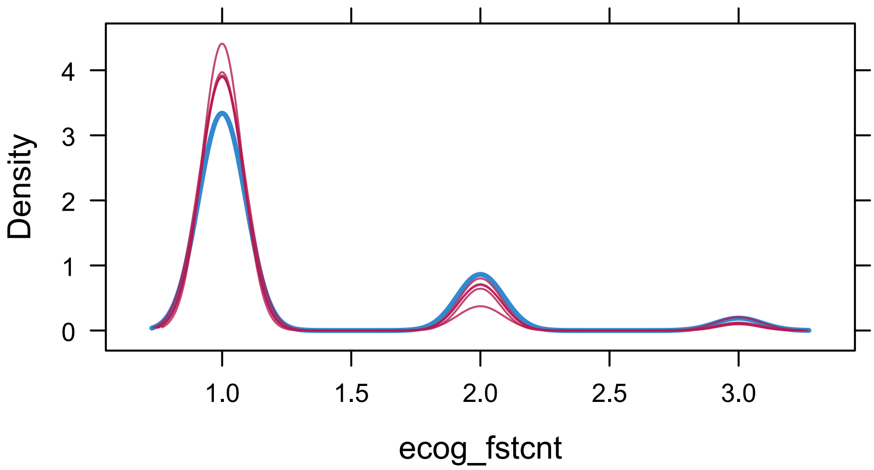


Figure 23: Density plot for ECOG performance status.

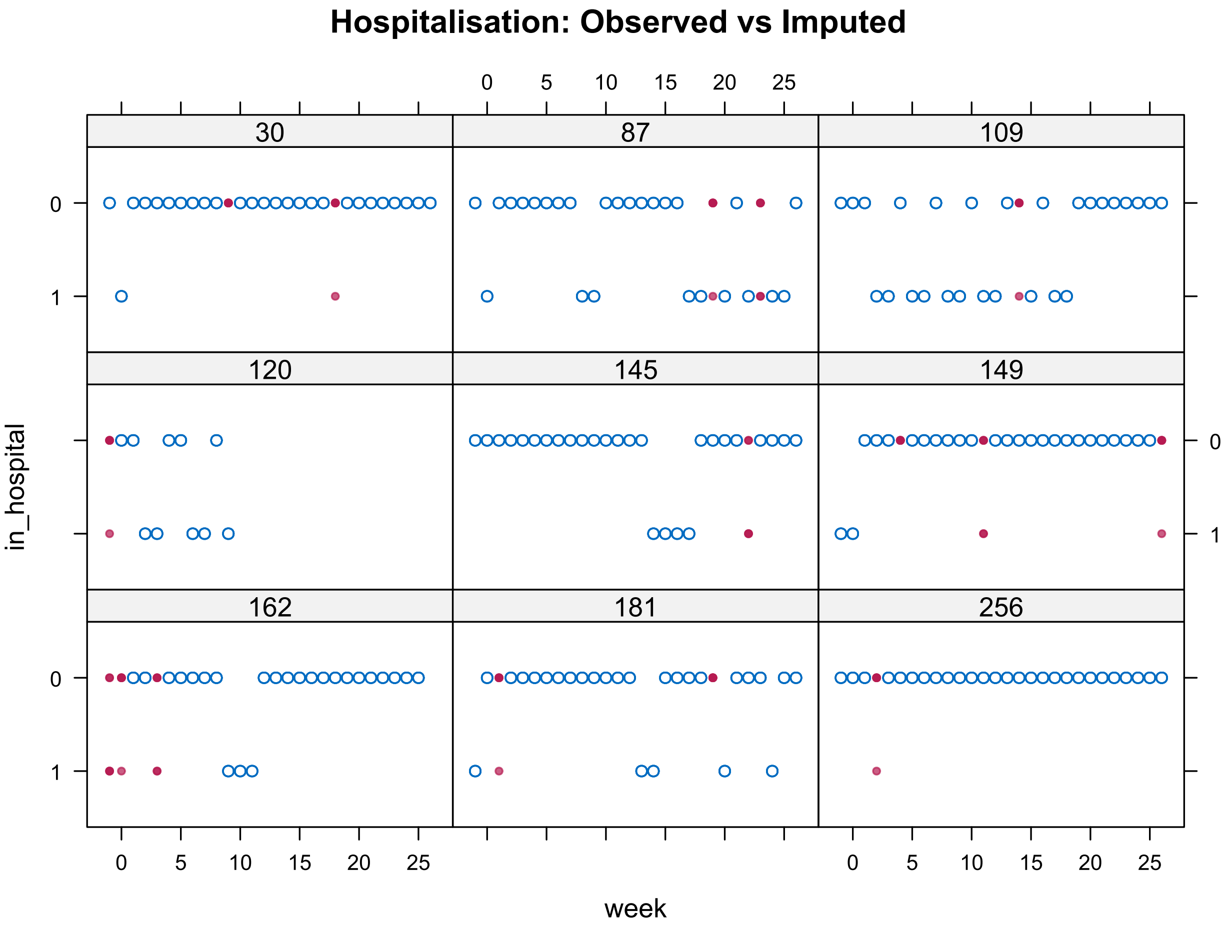


Figure 24: Scatter plots of imputed binary indicators against time, by imputed dataset.

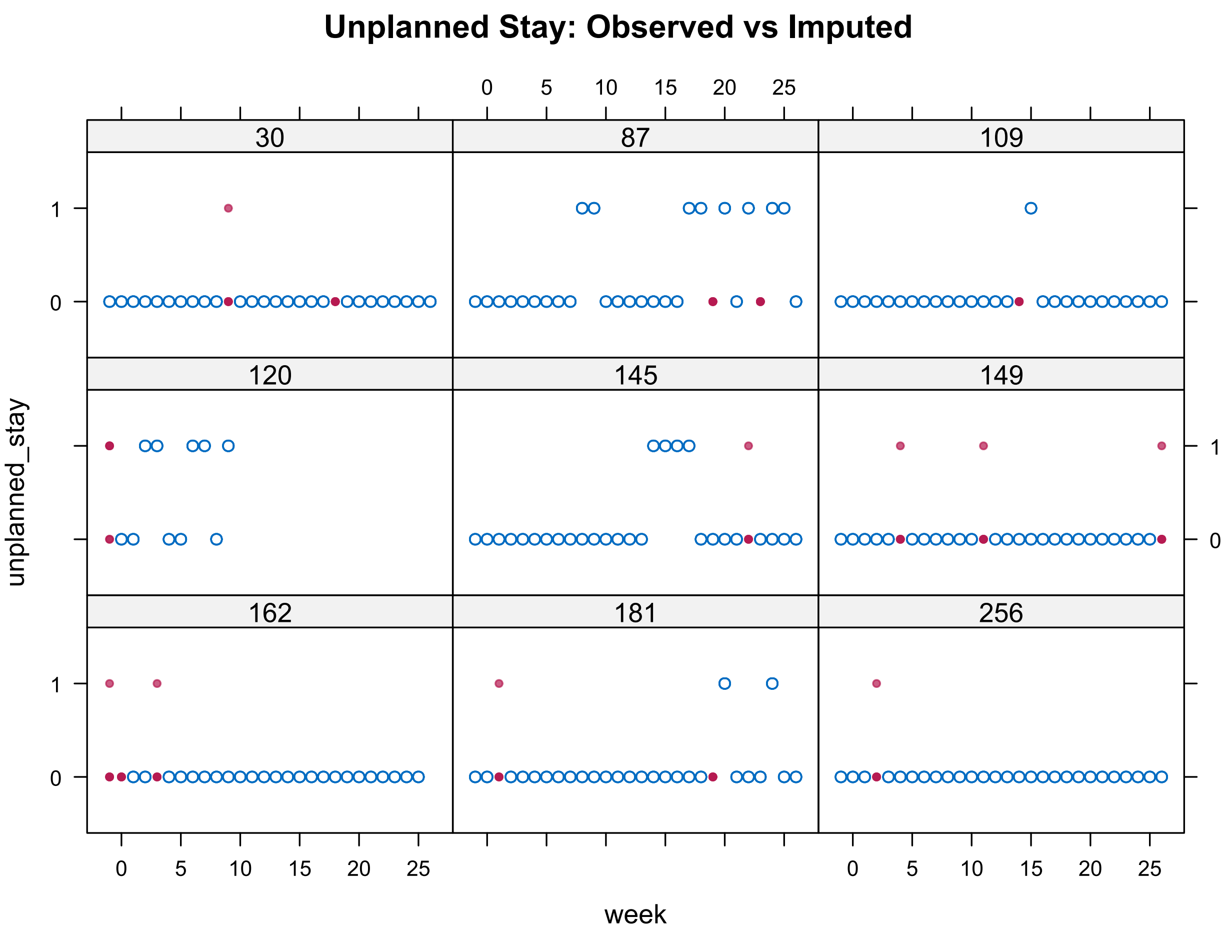


Figure 25: Scatter plots of imputed binary indicators against time, by imputed dataset.

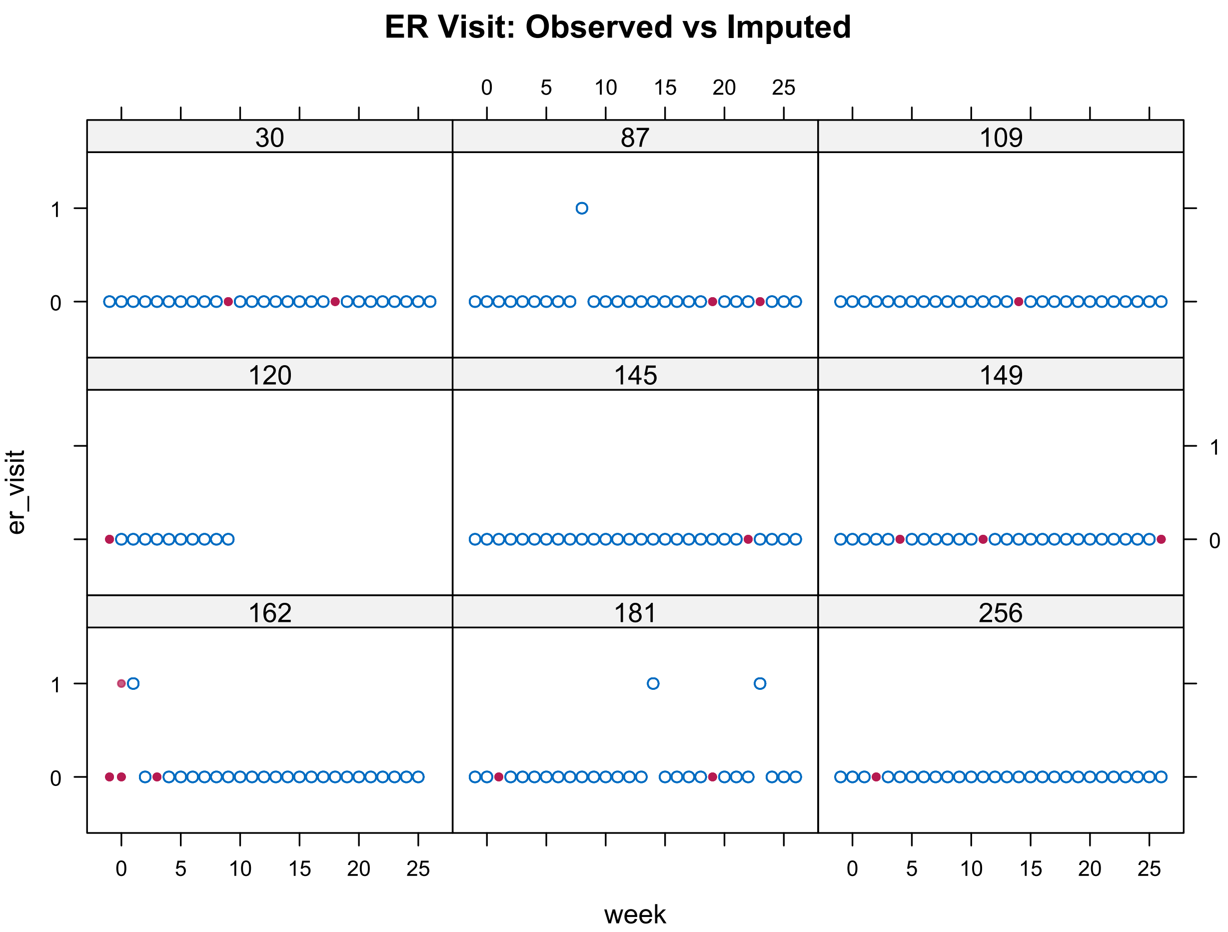


Figure 26: Scatter plots of imputed binary indicators against time, by imputed dataset.

13.3.8 Constructing the Ordinal TAOOH Score

After imputation of the component binary indicators, we construct a 5-level ordinal score representing the patient's weekly health state. The ordinal score is derived deterministically from the imputed indicators according to the following hierarchy:

Score	State	Condition
5	Dead	`death == 1`
4	Unplanned Hospitalisation	`in_hospital == 1` and `unplanned_stay == 1`
3	Emergency Room Visit	`er_visit == 1`
2	Planned Hospitalisation	`in_hospital == 1` and `unplanned_stay == 0`
1	Alive and Out of Hospital	Otherwise

Table 13: Definition of the 5-level ordinal TAOOH score

13.3.9 Analysis of Trial Data

For the final trial analysis, all $m = 50$ imputed datasets will be analysed, and posterior draws will be stacked to propagate imputation uncertainty into the final inference.

13.4 Multiple Imputation Strategy for Overall Survival

13.4.1 Overview

This appendix describes the multiple imputation strategy for the Overall Survival (OS) outcome in the LIQPLAT trial. The approach uses Multiple Imputation by Chained Equations (MICE) to handle missing baseline covariate data under the Missing At Random (MAR) assumption.

Since OS data are structured in wide format (one row per patient), we use standard single-level imputation methods appropriate for each variable type.

13.4.2 Data

The following code uses historical data from the Division of Medical Oncology at USB (first appointments from 2023 to mid-2024, prior to LIQPLAT initiation). These data are used to develop and validate the imputation procedure.

13.4.3 Data Preprocessing

13.4.3.1 Variable Transformations

Survival time is censored at 182 days (6 months) for the primary analysis. The death event indicator is derived accordingly.

13.4.3.2 Nelson-Aalen Cumulative Hazard Estimate

We include the Nelson-Aalen cumulative hazard estimate (`na_est`) as a predictor in the imputation model instead of survival time (Buuren 2021, chapter 9.1).

13.4.4 Imputation Variables

Following Mathur, Zhang, and Shpitser (2025), we restrict the imputation model to variables included in the analysis model or those acting causally on the incomplete variables or their missingness indicators.

13.4.4.1 Analysis Variables

Variable	Description	Type
`tx`	Treatment group allocation	Binary
`ecog_fstcnt`	ECOG performance status at baseline (0–4)	Ordered factor
`stage`	Tumor stage (locally advanced vs metastasized)	Binary
`albumin`	Serum albumin at baseline (g/L)	Continuous
`c_reactive_protein`	C-reactive protein at baseline (mg/L)	Continuous
`event_death`	Death indicator (within 182 days)	Binary

Table 14: Analysis variables included in the imputation model

13.4.4.2 Auxiliary Variables

Variable	Description	Type
`age`	Age at date of randomisation (years)	Continuous
`gender`	Patient sex	Binary
`bilirubin`	Serum bilirubin at baseline (μmol/L)	Continuous
`lactate_dehydrogenase`	Lactate dehydrogenase at baseline (U/L)	Continuous
`na_est`	Nelson-Aalen cumulative hazard estimate	Continuous

Table 15: Auxiliary variables included in the imputation model

13.4.5 Multiple Imputation Procedure

13.4.5.1 Imputation Methods

Variable	Method	Description
`ecog_fstcnt`	`polr`	Proportional odds logistic regression for ordered factors
`stage`	`logreg`	Logistic regression for binary variables
`albumin`	`pmm`	Predictive mean matching for continuous variables
`bilirubin`	`pmm`	Predictive mean matching for continuous variables
`c_reactive_protein`	`pmm`	Predictive mean matching for continuous variables
`lactate_dehydrogenase`	`pmm`	Predictive mean matching for continuous variables

Table 16: Imputation methods by variable

Predictive mean matching (PMM) is used for continuous variables to preserve the marginal distributions and avoid imputing implausible values. We use Type 1 PMM with a donor pool of $k = 5$ (Buuren 2021, chapter 3.4). For ordered categorical variables (ECOG), proportional odds logistic regression (`polr`) maintains the ordinal structure.

13.4.5.2 Predictor Matrix Configuration

The predictor matrix is configured to:

- Exclude `pat_id` from predictions (identifier only)
- Exclude `survival_time` from predictions (we use `na_est` instead to incorporate survival information)
- Allow all other variables to predict each other

```r
# Note: We don't have stage well defined in historic data, so we will use diagnosis (5
level factor) instead for illustrative purposes
md.pattern(df)
```

```r
init <- mice(df, maxit = 0)
meth <- init$method
pred_matrix <- init$predictorMatrix

# Patient ID should not be used for imputation
pred_matrix[, "pat_id"] <- 0
pred_matrix["pat_id", ] <- 0
meth["pat_id"] <- ""

# Use Nelson-Aalen estimate instead of raw survival time
pred_matrix[, "survival_time"] <- 0
pred_matrix["survival_time", ] <- 0
meth["survival_time"] <- ""

imputed_data <- mice(df,
                     m = 5,       # 50 in final analysis
                     maxit = 5,  # 50 in final analysis
                     predictorMatrix = pred_matrix,
                     method = meth,
```

```
                    donors = 5,
                    seed = 1234)
```

13.4.5.3 Imputation Parameters

For the final trial analysis:

- Number of imputed datasets: $m = 50$
- Maximum iterations: 50
- Donor pool size: $k = 5$

13.4.6 Convergence Diagnostics

Convergence is assessed via trace plots of the mean and standard deviation of imputed values across iterations.

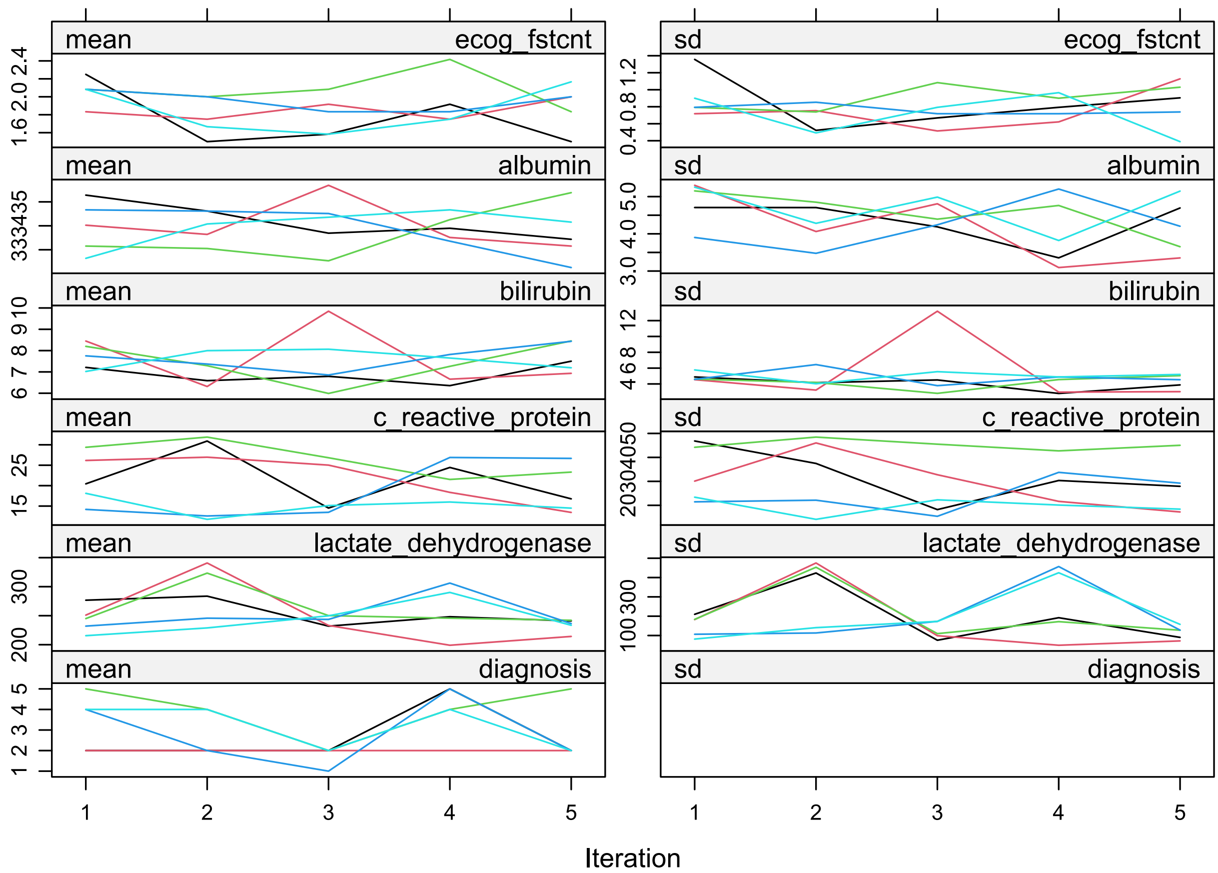


Figure 27: Trace plots showing the mean and standard deviation of imputed values across iterations for each imputed dataset.

13.4.7 Distribution Diagnostics

We compare the distributions of observed and imputed values to assess plausibility.

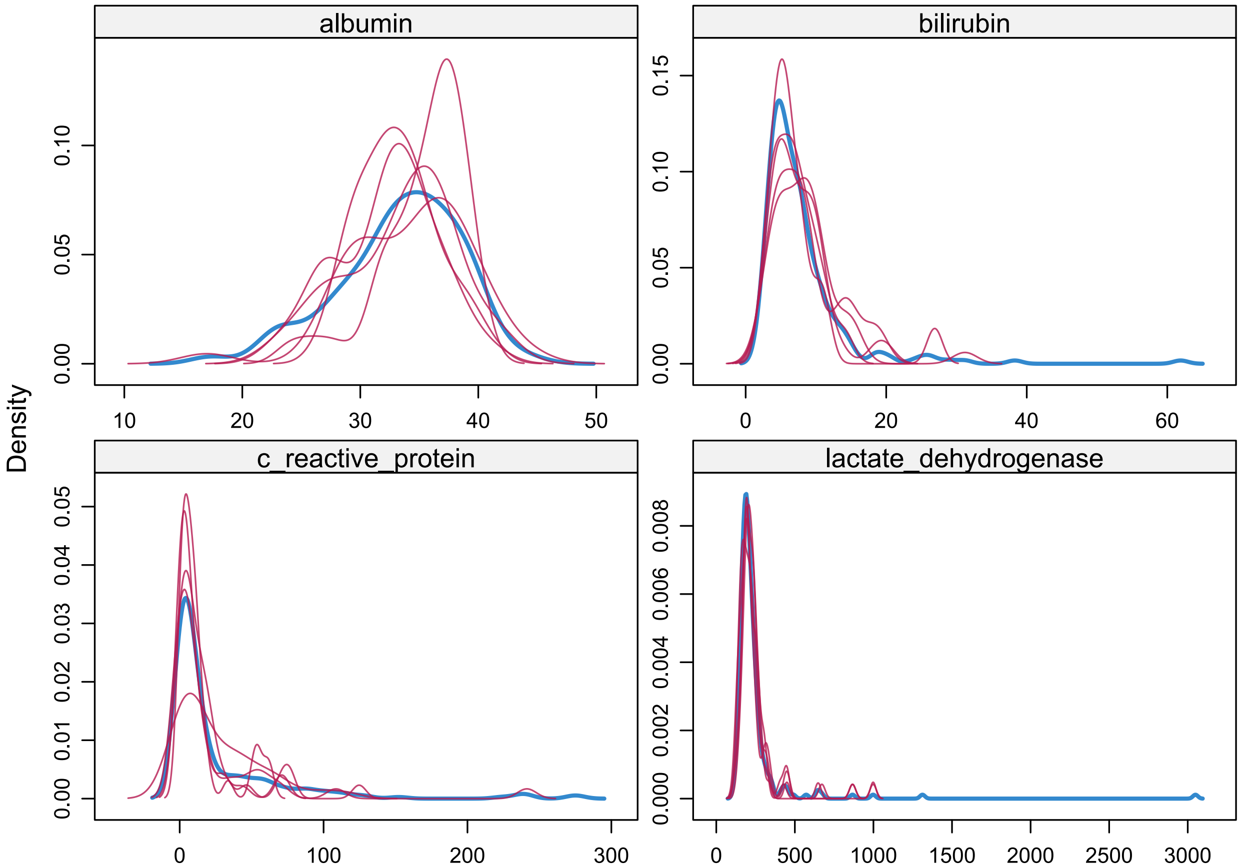


Figure 28: Density plots comparing observed (blue) and imputed (red) values for all imputed variables.

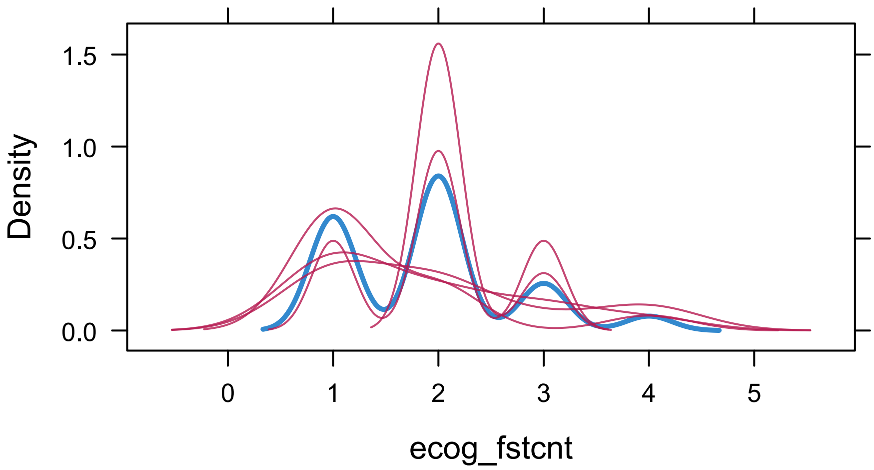


Figure 29: Density plot for ECOG performance status.

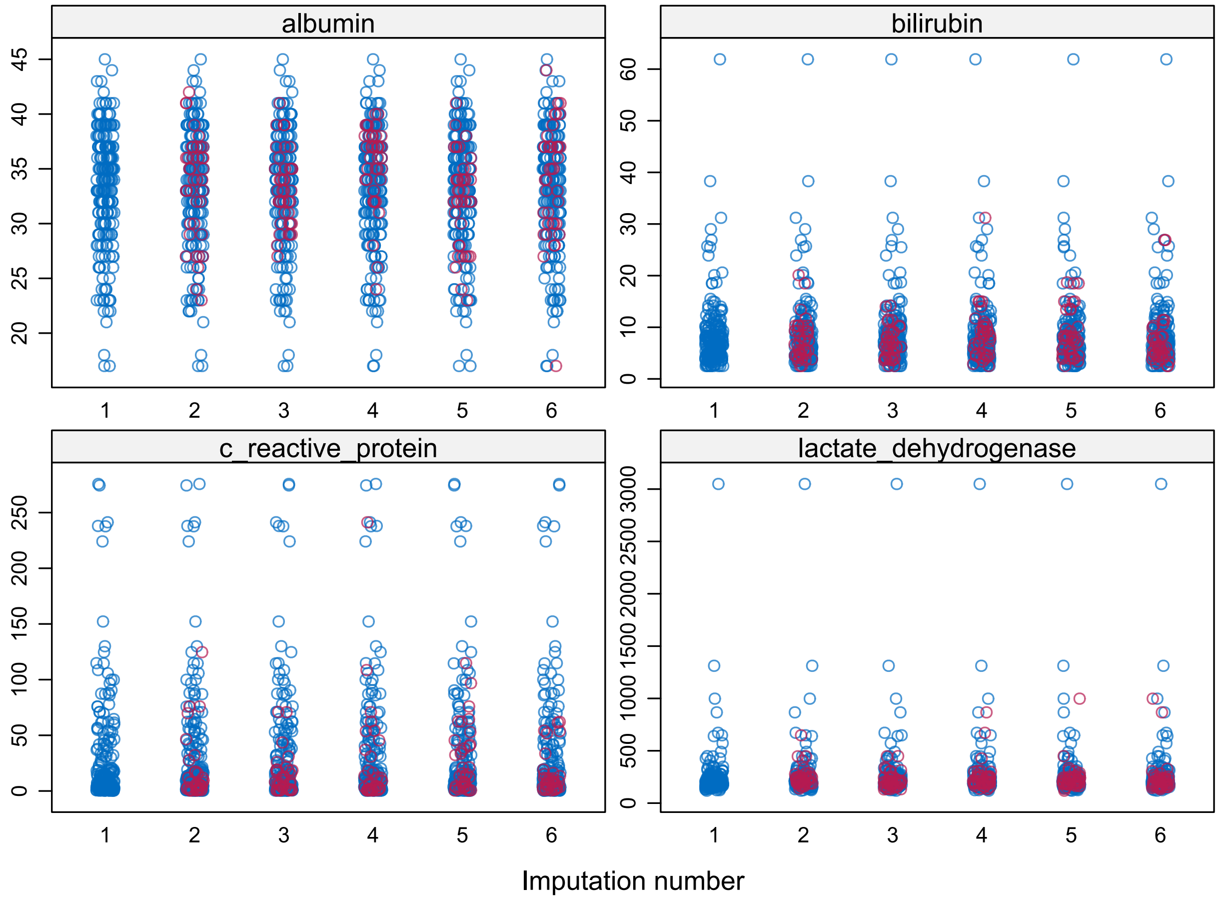


Figure 30: Strip plots of imputed values for continuous laboratory variables across imputed datasets.

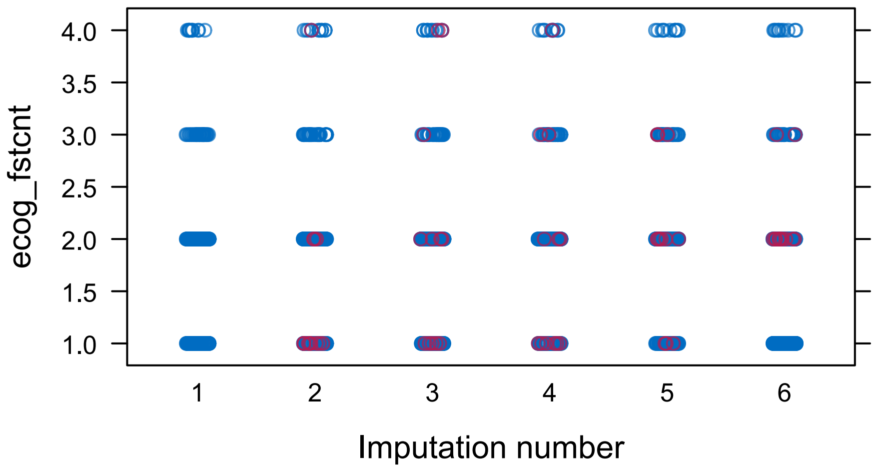


Figure 31: Strip plot of imputed values for ECOG performance status across imputed datasets.

13.4.8 Analysis of Trial Data

For the final trial analysis, all $m = 50$ imputed datasets will be analysed, and posterior draws will be stacked to propagate imputation uncertainty into the final inference.

13.5 QoL Interval Censoring Investigation

13.5.1 Background

An alternative to the Principal Stratum analysis for QoL is to model the 8-state process (7 QoL states + Death) directly, treating unobserved weeks for living patients as interval-censored (state is in `[1, 7]`). The `rmsb` package supports this via `Ocens()`.

13.5.2 Method

We tested this using a simulated dataset derived from historical data, where the true state trajectories were known. We created a sparse dataset (15% observations retained post-baseline) and fitted a first-order Markov model using `Ocens(y.a, y.b)` where `y.a=1, y.b=7` for missing weeks of alive patients, and `y.a=y.b=8` at death.

13.5.3 Findings

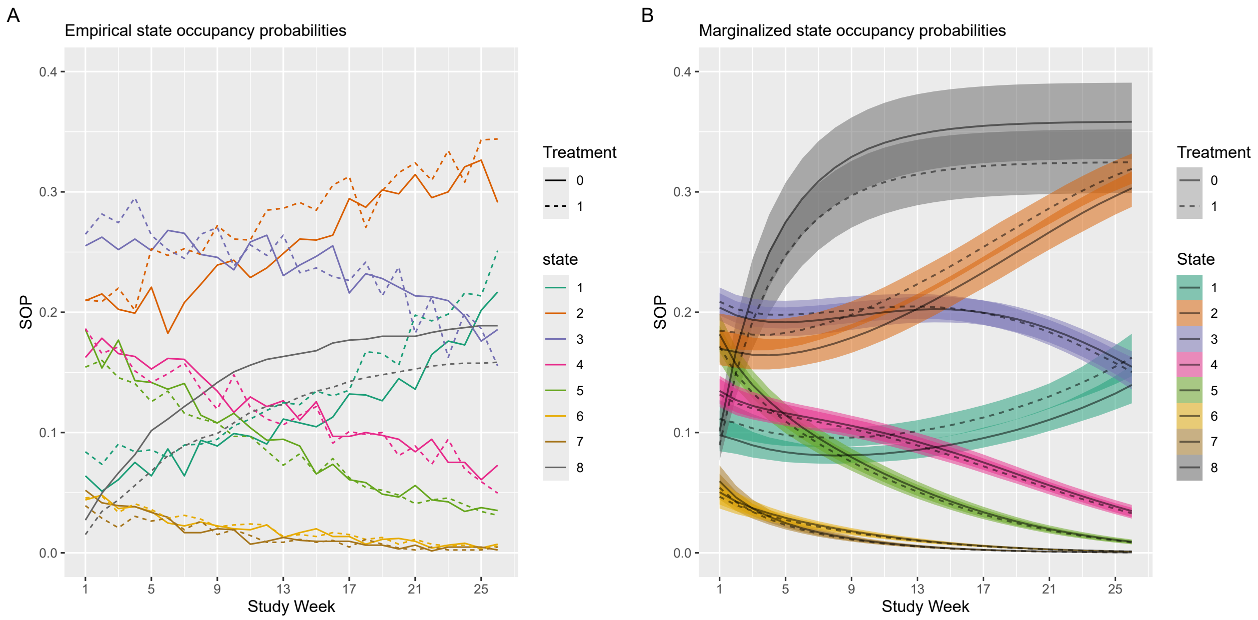


Figure 32: Comparison of empirical SOPs (A) vs. model-derived SOPs using interval censoring (B). The model significantly overestimates the cumulative incidence of death (State 8).

The interval censoring approach led to a substantial overestimation of the cumulative incidence of death (Figure 32). This bias likely arises because the absorbing death state is always observed precisely, while the intermediate living states are heavily censored. The model appears to incorrectly assign probability mass over time towards the known absorbing state. This bias propagates, underestimating time spent in living states.

13.5.4 Conclusion

Due to this potential for significant bias with sparse intermediate data, the interval censoring approach was deemed unsuitable for the primary QoL analysis, leading to the adoption of the Principal Stratum approach with its specific assumptions.

13.6 TAOOH State Occupancy Probabilities

The state occupancy probability (SOP) is a foundational derived quantity from a transition model. For a second-order Markov process, the SOP is the probability of being in state l at time t, conditional on baseline covariates $\mathbf{X}$ and two initial states observed prior to the first prediction. Let Y_{-1} and Y_0 denote the states at times -1 and 0, respectively. The SOP is then $S_t(l) = P(Y_t = l \mid \mathbf{X}, Y_{-1}, Y_0)$ for $t \geq 1$.

In a first-order model, a single transition matrix is sufficient to describe the evolution of the system from time $t-1$ to t (Rohde et al. 2024). In the second-order case, the probability of transitioning to state l at time t depends on both the state at $t-1$ and the state at $t-2$. This dependency requires a set of transition matrices —one for each possible prior state $Y_{t-2} = h$—rather than a single matrix. The state of the system at any given time is described by the joint probability distribution of the last two outcomes. Therefore, to compute the SOPs, we must recursively calculate the evolution of this joint probability distribution over time.

Let the conditional transition probability at time t be denoted as $T_t(l \mid h, k) = P(Y_t = l \mid Y_{t-2} = h, Y_{t-1} = k, \mathbf{X})$. These probabilities, which represent the chance of moving to state l given the history of being in state h then state k, are calculated directly from the fitted ordinal transition model for a specific covariate profile $\mathbf{X}$. We denote the joint probability of consecutive states ending at time t as $J_t(k, l) = P(Y_{t-1} = k, Y_t = l)$.

The unconditionalization process proceeds as follows:

1. **Initialization**: We begin with the known initial states Y_{-1} and Y_0. The joint probability $J_0(h, k) = P(Y_{-1} = h, Y_0 = k)$ is initialized as a matrix of zeros with a 1 at the position $(h, k) = (Y_{-1}, Y_0)$. The initial SOP at $t = 0$ is known with certainty: $S_0(l) = 1$ for state $l = Y_0$ and 0 otherwise.

2. **Recursive Calculation of Joint Probabilities**: For each subsequent time point $t = 1, 2, ..., d$, we calculate the new joint probability distribution $J_t(k, l) = P(Y_{t-1} = k, Y_t = l)$ by marginalizing over the state at time $t-2$, denoted by h. Using the law of total probability, the recursion is:

$$J_t(k, l) = \sum_{h=1}^{K} T_t(l \mid h, k) \times J_{t-1}(h, k)$$

   Here, $J_{t-1}(h, k) = P(Y_{t-2} = h, Y_{t-1} = k)$ provides the joint distribution at the previous step, and $T_t(l \mid h, k)$ gives the transition probability to state l. Their product, summed over all possible values of h, yields the updated joint distribution $J_t(k, l)$.

3. **Calculation of State Occupancy Probabilities**: Once the joint probability matrix J_t for time t is obtained, the SOP for being in state l at time t, denoted $S_t(l)$, is calculated by marginalizing out the state at time $t-1$:

$$S_t(l) = P(Y_t = l) = \sum_{k=1}^{K} J_t(k, l)$$

This $S_t(l)$ is the marginal probability of being in state l at time t, conditional on the baseline covariates and initial states: $P(Y_t = l \mid \mathbf{X}, Y_{-1}, Y_0)$. This three-step process is repeated for all time points up to the desired horizon d, yielding the full set of SOPs for a subject with a given covariate profile and specific initial history. The code that implements these calculations can be found here.

13.7 Overall Survival Implementation Code

This appendix provides example R code for the Overall Survival analysis described in the main text.

13.7.1 Model Fitting

```r
# Define priors
tx_scale <- 0.55
default_scale <- 2.5

prior <- normal(
  location = c(rep(0, 5)),
  scale = c(tx_scale, rep(default_scale, 4))
)

# Fit the model
model_os <- stan_surv(
    formula = Surv(day, death) ~ tx + ecog_binary + stage_binary +
              mgps,
    data = data,
    basehaz = "ms",
    basehaz_ops = list(df = 5),
    prior = prior,
    prior_intercept = normal(0, 20),
    prior_aux = dirichlet(1),
    chains = 4,
    cores = 4,
    seed = 68818,
    iter = 4000
)
```

13.7.2 RMST Calculation

```r
# --- Calculate RMST ---
# Get posterior survival curves for tx=0
ps_0_list <- posterior_survfit(
    model,
    newdata = data_for_model |> mutate(tx = 0),
    times = 0,
    extrapolate = TRUE,
    standardise = TRUE,
    control = list(edist = 182, epoints = 200),
    return_matrix = TRUE,
    draws = 4000
)

std_surv_matrix_0 <- do.call(cbind, ps_0_list)
time_points <- sapply(ps_0_list, attr, "times")

# Get posterior survival curves for tx=1
ps_1_list <- posterior_survfit(
    model,
    newdata = data_for_model |> mutate(tx = 1),
    times = 0,
    extrapolate = TRUE,
    standardise = TRUE,
```

```r
    control = list(edist = 182, epoints = 200),
    return_matrix = TRUE,
    draws = 4000
)
std_surv_matrix_1 <- do.call(cbind, ps_1_list)

# Calculate RMST for tx=0 using trapezoidal rule
delta_t <- diff(time_points)
s_t1_0 <- std_surv_matrix_0[, -ncol(std_surv_matrix_0)]
s_t2_0 <- std_surv_matrix_0[, -1]
trapezoid_areas_0 <- 0.5 * (s_t1_0 + s_t2_0) *
    matrix(delta_t, nrow = nrow(s_t1_0), ncol = length(delta_t), byrow = TRUE)
rmst_0 <- rowSums(trapezoid_areas_0)

# Calculate RMST for tx=1
s_t1_1 <- std_surv_matrix_1[, -ncol(std_surv_matrix_1)]
s_t2_1 <- std_surv_matrix_1[, -1]
trapezoid_areas_1 <- 0.5 * (s_t1_1 + s_t2_1) *
    matrix(delta_t, nrow = nrow(s_t1_1), ncol = length(delta_t), byrow = TRUE)
rmst_1 <- rowSums(trapezoid_areas_1)

# Calculate the difference
rmst_diff <- rmst_1 - rmst_0

# Combine and save results
rmst_results <- data.frame(rmst_0, rmst_1, rmst_diff)
```

13.8 Quality of Life Implementation Code

This appendix provides example R code for the Quality of Life analysis described in the main text.

13.8.1 Model Fitting

```
# Example R code for model fitting
pcontrast_arg <- list(
  c1 = list(tx = 1),
  c2 = list(tx = 0),
  contrast = expression(c1 - c2),
  mu = 0,
  sd = 0.5
)

model_qol <- blrm(
  formula = y ~ tx + rcs(week, 4) + yprev * gap + ecog_fstcnt + diagnosis,
  pcontrast = pcontrast_arg,
  data = data_survivors,
  ppo = ~week,             # Allow week effect to be non-proportional
  cppo = function(y) y,    # Constraint: non-proportionality linear in y
  iter = 4000, chains = 4, seed = 68818
)
```

13.8.2 Deriving State Occupancy Probabilities

```
baseline_df <- data_for_model |>
    group_by(id) |>
    arrange(time) |>
    slice_head(n = 1) |>
    ungroup() |>
    mutate(
        time = 1,
        gap = 1,
        y = NA) |>
    as.data.table()
```

```
write_SOP <- function(i, tx, baseline_df, path){

  row <- baseline_df[i]

  sops <-
  soprobMarkovOrdm(
    model,
    data = list(tx = tx, yprev=row$yprev, gap = row$gap),
    times = 1:26,
    ylevels = 1:7,
    absorb = NULL,
    tvarname = "time",
    pvarname = "yprev"
  )
  sops <- sops[1:2000,,]

  sops <- as.data.table(sops)
```

```r
  # Rename columns
  setnames(sops,
           old = c("V1", "V2", "V3", "value"),
           new = c("draw", "time", "state", "sop"))

  sops$time <- as.integer(sops$time)
  sops$state <- as.factor(sops$state)
  sops$tx <- tx
  sops$i <- i

  folder <- file.path(path, glue::glue("marginalized_sop_{tx}"),
glue::glue("msop_{i}.parquet"))

  # Create the directory if it doesn't exist
  dir_to_create <- dirname(folder)
  if (!dir.exists(dir_to_create)) {
    dir.create(dir_to_create, recursive = TRUE)
  }

  arrow::write_parquet(
    x = sops,
    sink = folder
    )
}
```

```r
path <- here::here("1-qol", "output", "primary-estimand")
future::plan(future::multisession, workers=6)


# Control
if (generate_SOPS) {
    furrr::future_walk(1:nrow(baseline_df), \(x) write_SOP(x, 0, baseline_df, path),
                       .progress=TRUE,
                       .options = furrr_options(seed=TRUE))
}
# Treatment
if (generate_SOPS) {
    furrr::future_walk(1:nrow(baseline_df), \(x) write_SOP(x, 1, baseline_df, path),
                       .progress=TRUE,
                       .options = furrr_options(seed=TRUE))
}
```

13.9 Time Alive and Out of Hospital Implementation Code

This appendix provides example R code for the Time Alive and Out of Hospital analysis described in the main text.

13.9.1 Model Fitting

```
pcontrast_arg <- list(
  c1 = list(tx = 1),
  c2 = list(tx = 0),
  contrast = expression(c1 - c2),
  mu = 0,
  sd = 0.541
)

model_taooh <- blrm(
    formula = y ~ tx + rcs(week, 4) + yprev + ypprev +
                week %ia% yprev + week %ia% ypprev +
                ecog_fstcnt + diagnosis + rcs(albumin, 3) +
                rcs(c_reactive_protein, 3) + cluster(id),
    pcontrast = pcontrast_arg,
    data = data_toh,
    iter = 4000, chains = 4, seed = 68818,
)
```

13.9.2 Deriving State Occupancy Probabilities

13.9.2.1 Predictions conditional on random effects

The predict function from rmsb ignores random effects.

```
predict_blrm_manual <- function(fit, newdata, include_re = FALSE) {
  if (!is.data.frame(newdata)) {
    stop("`newdata` must be a data frame.")
  }
  if (include_re && !"id" %in% colnames(newdata)) {
    stop("`newdata` must contain a 'id' column when include_re=TRUE")
  }

  N_new <- nrow(newdata)

  # --- 1) Get the design matrix for the new data ---
  X <- rms::predictrms(fit, newdata = newdata, type = 'x')

  # --- 2) Extract posterior draws and model information ---
  draws <- fit$draws
  ndraws <- nrow(draws)
  ns <- fit$non.slopes
  K <- ns + 1

  intercept_draws <- draws[, 1:ns, drop = FALSE]
  beta_draws <- draws[, (ns + 1):ncol(draws), drop = FALSE]

  # --- 3) Calculate the fixed-effect linear predictor (eta_fixed) ---
  eta_fixed <- beta_draws %*% t(X)

  # --- 4) Get the random effect draws (u_draws) if requested ---
```

```
  u_draws <- matrix(0, nrow = ndraws, ncol = N_new)

  if (include_re) {
    if (length(unique(newdata$id)) > 1) {
      stop("`newdata` contains multiple unique patient IDs. This version is restricted
to one at a time.")
    }
    if (!requireNamespace("posterior", quietly = TRUE)) {
      stop("The 'posterior' package is required to extract random effect draws.")
    }
    sf <- rmsb::stanGet(fit)
    all_gamma_draws <- posterior::as_draws_matrix(sf$draws(variables = "gamma"))
    patient_ids <- newdata$id
    u_draws <- all_gamma_draws[, patient_ids, drop = FALSE]
  }

  # --- 5) Calculate the total linear predictor (eta) ---
  stopifnot(all(dim(eta_fixed) == dim(u_draws)))
  eta <- eta_fixed + u_draws

  # --- 6) Calculate probabilities for each observation and each draw ---
  p_cat_draws_list <- vector("list", N_new)

  for (i in 1:N_new) {
    eta_i <- eta[, i]
    s_ge <- plogis(sweep(intercept_draws, 1, eta_i, "+"))

    p_cat_draws <- matrix(NA_real_, nrow = ndraws, ncol = K)
    p_cat_draws[, 1] <- 1 - s_ge[, 1]
    if (K > 2) {
      for (kk in 2:(K - 1)) {
        p_cat_draws[, kk] <- s_ge[, kk - 1] - s_ge[, kk]
      }
    }
    p_cat_draws[, K] <- s_ge[, K - 1]
    colnames(p_cat_draws) <- paste0("y=", 1:K)

    p_cat_draws_list[[i]] <- p_cat_draws
  }

  # --- 7) Reshape draws into a single 3D array ---
  draws_array <- array(
    data = NA_real_,
    dim = c(ndraws, N_new, K)
  )

  for (i in 1:N_new) {
    draws_array[, i, ] <- p_cat_draws_list[[i]]
  }

  dimnames(draws_array) <- list(
    NULL,
    rownames(newdata),
    colnames(p_cat_draws_list[[1]])
  )
```

```
  return(draws_array)
}
```

13.9.2.2 Unconditionalizing Predictions

```
#' Calculate Second-Order State Occupation Probabilities
#'
#' This function calculates the unconditional state occupation probabilities for a
#' second-order Markov model.
#'
#' @param model_fit The fitted model object (rmsb or brms)
#' @param baseline_data A single-row data frame containing the initial states and
#'   all baseline predictor values required by the model.
#' @param col_yprev A string with the column name for the state at t-1 (e.g., "yprev").
#' @param col_ypprev A string with the column name for the state at t-2 (e.g.,
"ypprev").
#' @param col_time A string with the column name that represents time in the model
(e.g., "week").
#' @param times A numeric vector specifying the time points for calculation (e.g.,
`1:26`).
#' @param n_states An integer for the number of levels in the ordinal outcome.
#' @param absorbing_states A numeric vector listing the absorbing states.
#'
#' @return A list containing two arrays:
#'   \item{unconditional}{A 3D array `(draws, states, time)` with the unconditional
state probabilities.}
#' @export

sop_markov_second_order <- function(model_fit,
                                    baseline_data,
                                    col_yprev,
                                    col_ypprev,
                                    col_time,
                                    times,
                                    n_states,
                                    absorbing_states,
                                    include_re = FALSE) {

  # --- Detect model type and get number of draws ---
  if (inherits(model_fit, "brmsfit")) {
    detected_type <- "brms"
    if (!requireNamespace("marginaleffects", quietly = TRUE) || !
requireNamespace("posterior", quietly = TRUE)) {
      stop("Packages 'marginaleffects' and 'posterior' are required for brms models.")
    }
    n_draws <- posterior::ndraws(model_fit)
  } else if (inherits(model_fit, "blrm")) {
    detected_type <- "rmsb"
    n_draws <- nrow(model_fit$draws)
  } else {
    stop("Unsupported model object class. Please provide a model from 'brms' (class
'brmsfit') or 'rmsb' (class 'blrm').")
  }
```

```r
# --- Parameters ---
M <- n_states
T_horizon <- max(times)

# --- Initial Conditions ---
initial_state_t_minus_1 <- as.numeric(baseline_data[[col_ypprev]])
initial_state_t_0 <- as.numeric(baseline_data[[col_yprev]])

# --- Data Structures to Store Results ---
P_joint_bayes <- array(0, dim = c(n_draws, M, M, T_horizon + 1),
                       dimnames = list(paste0("draw_", 1:n_draws),
                                       paste0("t-1=", 1:M),
                                       paste0("t=", 1:M),
                                       paste0("time=", 0:T_horizon)))

P_uncond_bayes <- array(0, dim = c(n_draws, M, T_horizon + 1),
                        dimnames = list(paste0("draw_", 1:n_draws),
                                        paste0("State_", 1:M),
                                        paste0("t=", 0:T_horizon)))

# Set the known probability at t=0
P_joint_bayes[, initial_state_t_minus_1, initial_state_t_0, 1] <- 1.0
P_uncond_bayes[, initial_state_t_0, 1] <- 1.0

# --- Create the static grid for batch predictions ---
history_grid <- expand.grid(
  h = factor(1:M, levels = 1:M), # X_{t-2}
  j = factor(1:M, levels = 1:M)  # X_{t-1}
)
names(history_grid) <- c(col_ypprev, col_yprev)

# --- Dynamically create the full prediction dataset ---
# Identify static covariates by excluding time-varying/history columns
cols_to_exclude <- c(col_yprev, col_ypprev, col_time)
static_covariate_names <- setdiff(names(baseline_data), cols_to_exclude)
static_covariates <- baseline_data[, static_covariate_names, drop = FALSE]

# Combine static covariates with the grid of all possible histories
prediction_data <- cbind(static_covariates, history_grid)

# --- Main Iteration Loop ---
for (t in times) {
  # Identify predictable histories
  predictable_idx <- which(
    !(prediction_data[[col_ypprev]] %in% absorbing_states) &
    !(prediction_data[[col_yprev]] %in% absorbing_states)
  )
  n_predictable <- length(predictable_idx)

  # Subset the data for prediction and set the current time
  prediction_data_subset <- prediction_data[predictable_idx, ]
  prediction_data_subset[[col_time]] <- t

  pred_array_3d <- NULL
```

```r
    if (detected_type == "brms") {
      # refactor previous states to remove absorbing states
      prediction_data_subset[[col_yprev]] <-
factor(prediction_data_subset[[col_yprev]], levels = c(1:M)[-absorbing_states])
      prediction_data_subset[[col_ypprev]] <-
factor(prediction_data_subset[[col_ypprev]], levels = c(1:M)[-absorbing_states])
      re_arg <- if (include_re) NULL else NA
      brms_pred_long <- marginaleffects::predictions(model_fit,
                                                     newdata = prediction_data_subset,
                                                     type = "response",
                                                     re_formula = re_arg) |>
                        marginaleffects::posterior_draws()

      pred_array_3d <- array(0, dim = c(n_draws, n_predictable, M))
      indices <- cbind(brms_pred_long$drawid,
                       brms_pred_long$rowid,
                       as.numeric(brms_pred_long$group))
      pred_array_3d[indices] <- brms_pred_long$draw

    } else { # detected_type == "rmsb"
    # dimension 1 = 4000 draws
    # dimension 2 = X_{t-1} and X_{t-2} combinations (rows from prediction_data_subset)
    # dimension 3 = cumulative probabilities (e.g., 4 columns for 5 states)
      pred_array_3d <- predict_blrm_manual(model_fit, prediction_data_subset,
include_re = include_re)
    }

    state_probs_all_draws <- do.call(rbind, lapply(1:dim(pred_array_3d)[1], function(d)
pred_array_3d[d, , ]))

    # --- Assemble transition probabilities for each draw ---
    # Create a 4D array to hold all transition probabilities: P(X_t=l | X_{t-2}=h,
X_{t-1}=j)
    # Dims: (draw, h, j, l)
    # So [, 1, 1, 1] give all draws for P(l = 1 | h = 1, j = 1)
    # [, 1, 1, ] for all draws for P(l | h = 1, j = 1) (rows must sum to 1)
    cond_probs_4d_array <- array(0, dim = c(n_draws, M, M, M))

    for (d in 1:n_draws) {
      # Create a placeholder matrix for this specific draw
      # each row is combination of (h, j), each column is l (rows must sum to 1)
      # this is for one draw only
      cond_probs_matrix_d <- matrix(0, nrow = M * M, ncol = M)

      # Handle absorbing states (P(j->j) = 1 if j is absorbing)
      absorbing_idx <- which(prediction_data[[col_yprev]] %in% absorbing_states)
      if (length(absorbing_idx) > 0) {
        absorbing_state <- as.numeric(as.character(prediction_data[[col_yprev]]
[absorbing_idx]))
        rows_and_cols <- cbind(absorbing_idx, absorbing_state)
        cond_probs_matrix_d[rows_and_cols] <- 1.0
      }

      # Get state probabilities for the current draw
      start_row <- (d - 1) * n_predictable + 1
```

```
      end_row <- d * n_predictable
      draw_d_probs <- state_probs_all_draws[start_row:end_row, ]

      # Place them into the correct rows
      cond_probs_matrix_d[predictable_idx, ] <- draw_d_probs

      # Reshape and store in the final 4D array
      cond_probs_4d_array[d, , , ] <- array(cond_probs_matrix_d, dim = c(M, M, M))
    }

    # --- Update Probabilities ---
    # Get joint probabilities from the previous step for all draws: P(X_{t-2}, X_{t-1})
    prev_P_joint_all_draws <- P_joint_bayes[, , , t]
    current_P_joint_all_draws <- array(0, dim = c(n_draws, M, M))

    for (d in 1:n_draws) {
      # This is P(X_{t-2}=h, X_{t-1}=j) for draw d
      prev_P_joint_d <- prev_P_joint_all_draws[d, , ]
      # P(X_t = l | X_{t-1} = j, X_{t-2} = h) for this draw
      cond_probs_d <- cond_probs_4d_array[d, , , ]

      # Law of Total Probability: P(X_t=l|h,j) * P(h,j)
      # weighted_probs <- cond_probs_d * array(prev_P_joint_d, dim = dim(cond_probs_d))
      weighted_probs <- sweep(cond_probs_d, c(1, 2), prev_P_joint_d, "*")
      # Marginalize over the t-2 state (h) to get P(X_{t-1}, X_t)
      # = Summation over h (the state at t-2), resulting in P(X_{t-1}=j, X_t=l)
      current_P_joint_d <- apply(weighted_probs, c(2, 3), sum)

      current_P_joint_all_draws[d, , ] <- current_P_joint_d

      # Marginalize over the t-1 state (j) to get P(X_t)
      uncond_probs_t <- colSums(current_P_joint_d)
      # Consider normalizing  to prevent floating point error accumulation (not doing
this for now)
      P_uncond_bayes[d, , t + 1] <- uncond_probs_t
      # P_uncond_bayes[d, , t + 1] <- uncond_probs_t / sum(uncond_probs_t)
    }

    P_joint_bayes[, , , t + 1] <- current_P_joint_all_draws
  }

  return(P_uncond_bayes)
}
```

13.9.3 Marginalization

```
write_SOP <- function(i, tx, baseline_df, path, model){

  row <- baseline_df[i, ]

  sops <-
  sop_markov_second_order(
    model_fit = model,
    baseline_data = data.frame(id = row$id, tx = tx, ecog_fstcnt = row$ecog_fstcnt,
diagnosis = row$diagnosis, albumin = row$albumin,  c_reactive_protein =
```

```
row$c_reactive_protein, yprev = row$yprev, ypprev = row$ypprev),
    col_yprev = "yprev",
    col_ypprev = "ypprev",
    col_time = "week",
    times = 1:26,
    n_states = 5,
    absorbing_states = c(5),
    include_re = TRUE
  )

  sops <- reshape2::melt(sops,
                   varnames = c("draw", "state", "time"),
                   value.name = "probability")
  sops <- sops |>
    mutate(
        time  = as.integer(sub("t=", "", time)),
        draw  = as.integer(sub("draw_", "", draw)),
        state = as.factor(sub("State_", "", state)),
        tx    = tx,
        i     = i ) |>
    filter(time > 0)

  folder <- file.path(path, glue::glue("/marginalized_sop_{tx}"), glue::glue("/
msop_{i}.parquet"))

  # Create the directory if it doesn't exist
  dir_to_create <- dirname(folder)
  if (!dir.exists(dir_to_create)) {
    dir.create(dir_to_create, recursive = TRUE)
  }

  arrow::write_parquet(
    x = sops,
    sink = folder
    )
}
```

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